



An Introduction to

# LARGE LANGUAGE MODELS

What it is and why it's important



## Structured Data

- .....> Tables, Spreadsheets, Databases
- .....> Format is predefined, easy to organise
- .....> Requires less storage, easier to manager and protect
- .....> **Small data problems:** Business value comes from collecting more data



## Unstructured Data

- .....> Images, audio, video, e-mails, knowledge bases
- .....> Native format, difficult to organise
- .....> Requires more storage, difficult to manage
- .....> **Big data problems:** Business value comes from interpreting the data at scale

# Big data problems

virtual assistant  
search  
content creation  
image classification  
text2design  
image reconstruction  
object detection  
machine translation  
object segmentation  
sentiment analysis

## Charateristics

Can be bootstrapped from existing solutions

Easy to collect data

Black box models

Initial business value is achieved quickly

## Small data problems

customer segmentation  
algorithmic trading  
weather prediction  
house price prediction  
churn prediction  
price elasticity  
fraud detection  
recommendation

### Charateristics

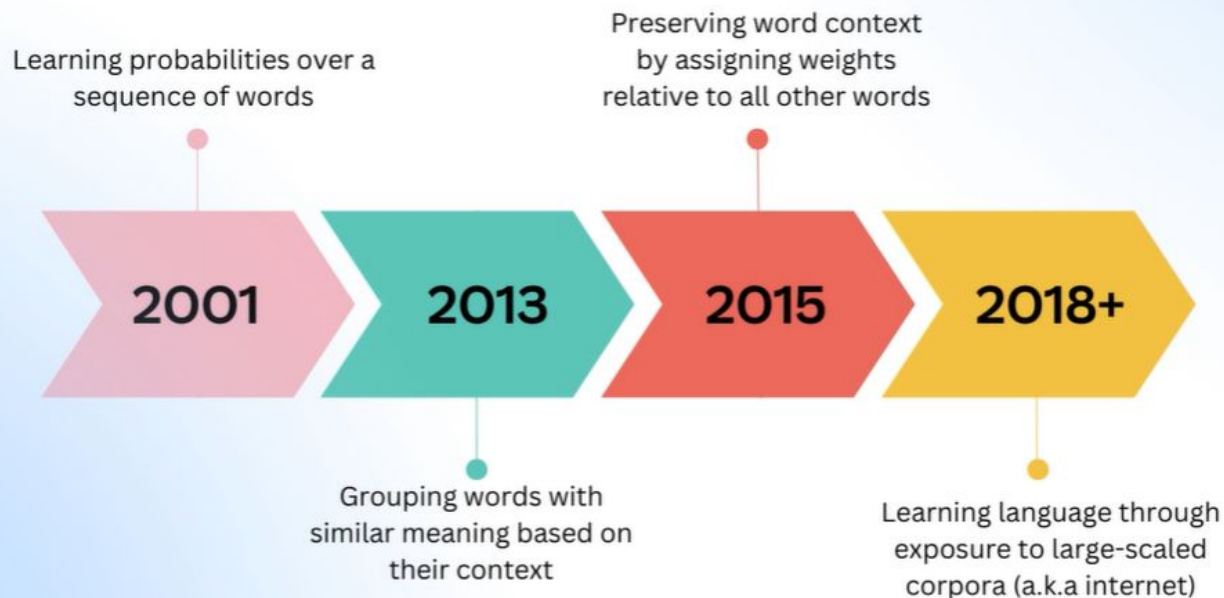
Requires deep business understanding

Data collection is a bottleneck

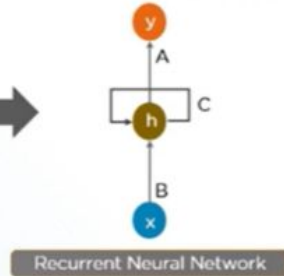
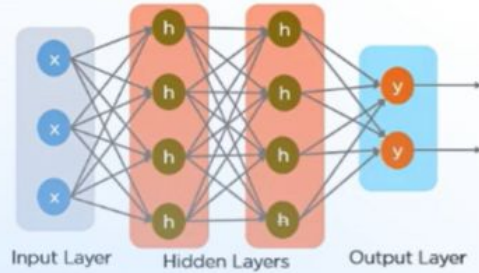
Models are (usually) interpretable

Delivers most of the business value

# The rise of Deep Learning Natural Language Processing



**The key idea was to create  
baseline model from where  
people can develop their  
specific use cases**



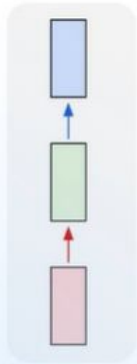
# Recurrent Neural Networks

RNN's were created to solve the lack of input memorization capabilities found in traditional Neural Networks

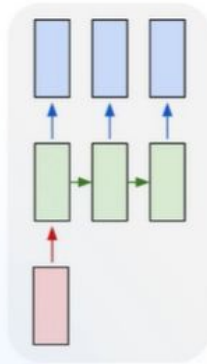
# RNN types

Architectures can be configured differently based on the type of inputs and outputs

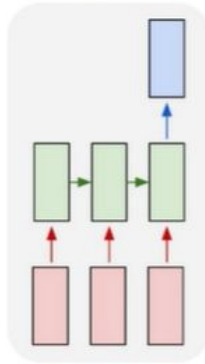
one to one



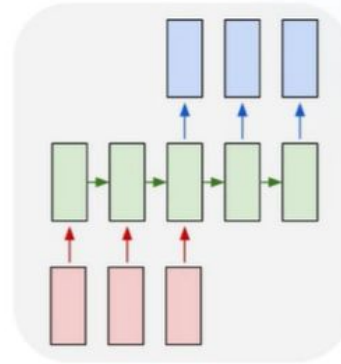
one to many



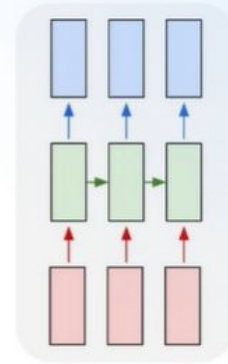
many to one



many to many

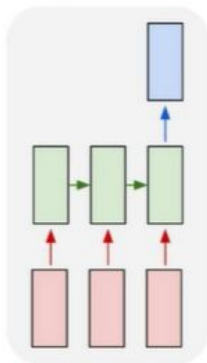


many to many



# Text Generation with RNN's

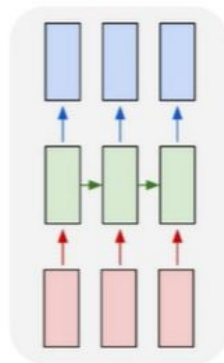
many to one



**One character at a time**

Sentiment Analysis

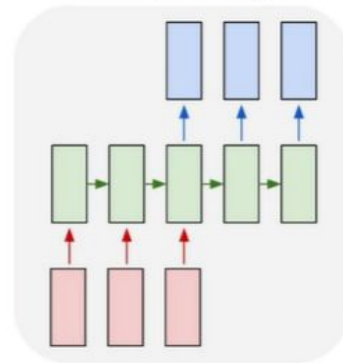
many to many



**Full Sentences (1)**

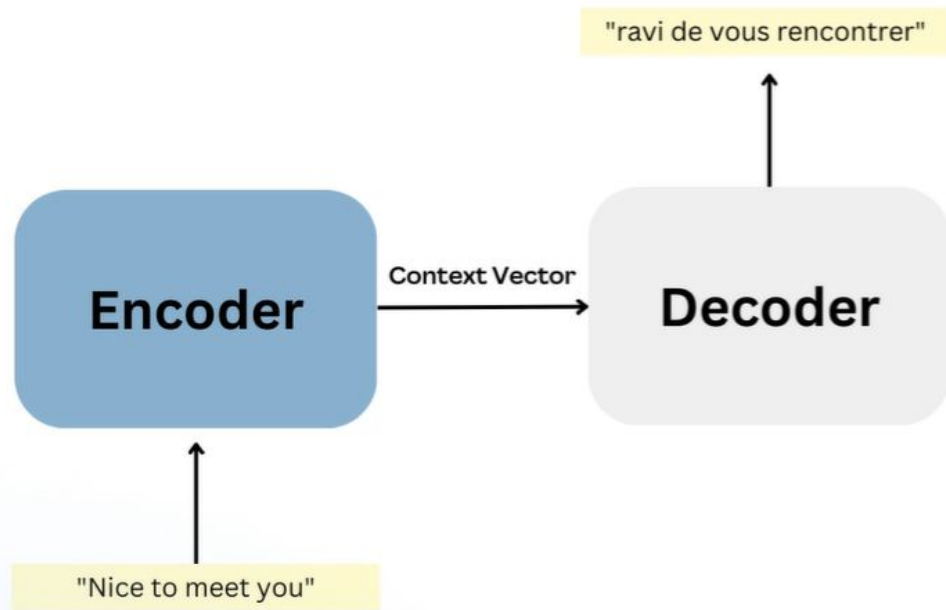
Machine Translation

many to many



**Full Sentences (2)**

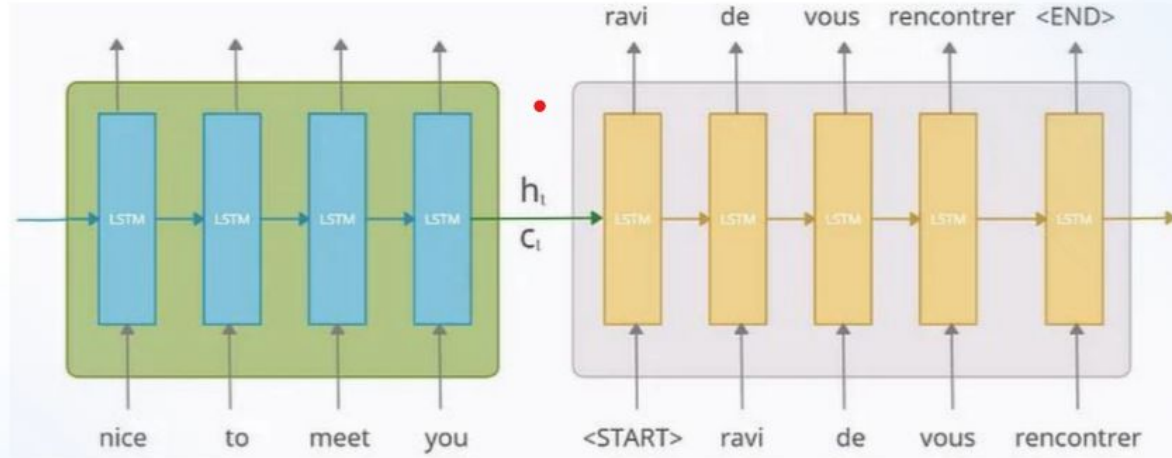
Summarization



# Encoder Decoder Architecture

Sets the standard for today's  
LLMs

## Encoder



## Decoder

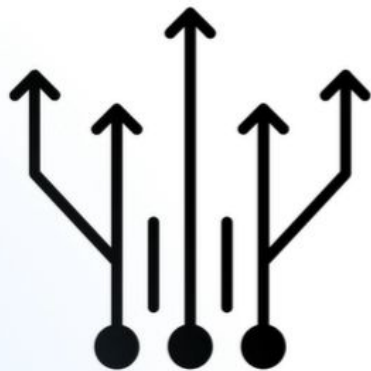
# What's under the hood?

# Machine Translation with RNNs

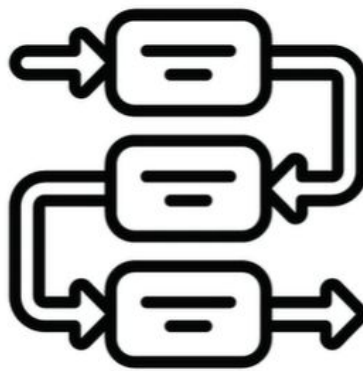


# but, what's wrong with RNNs?

Can't be trained  
in parallel



Long Range  
Dependency



Sequential  
Processing



# Language is complex, words can be ambiguous

The teacher's book?

The teacher taught the student with the book

The student's book

**we need more than word sequences..**



... we need understanding of the full context

# Attention is all you need!

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## Attention Is All You Need

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### Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.



# Transformers to the rescue!

Scale efficiently: Large datasets

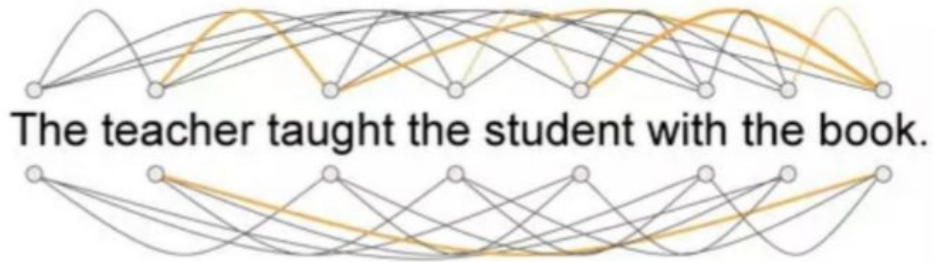
Easy to Parallelize

Better Contextual Learning

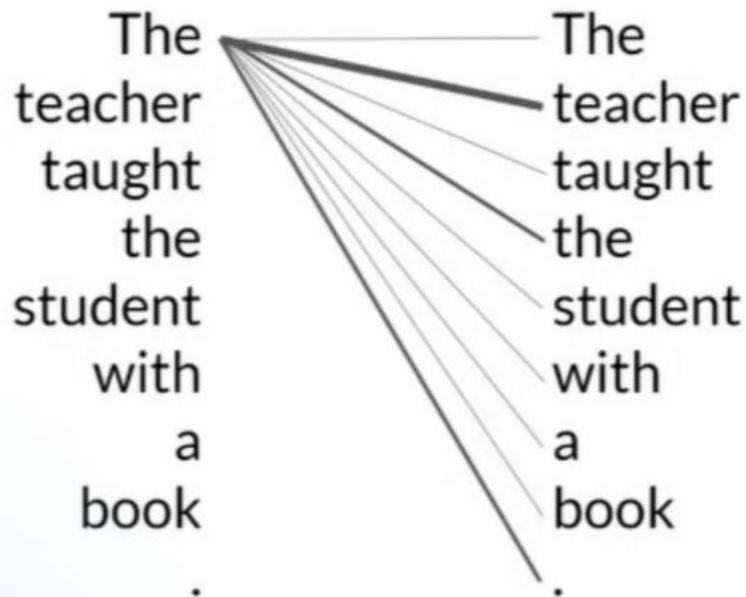




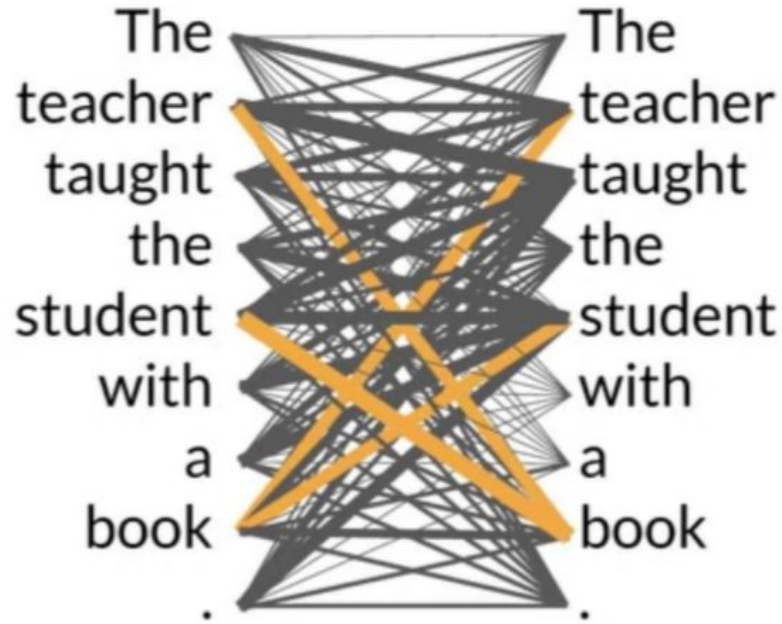
The teacher taught the student with the book.



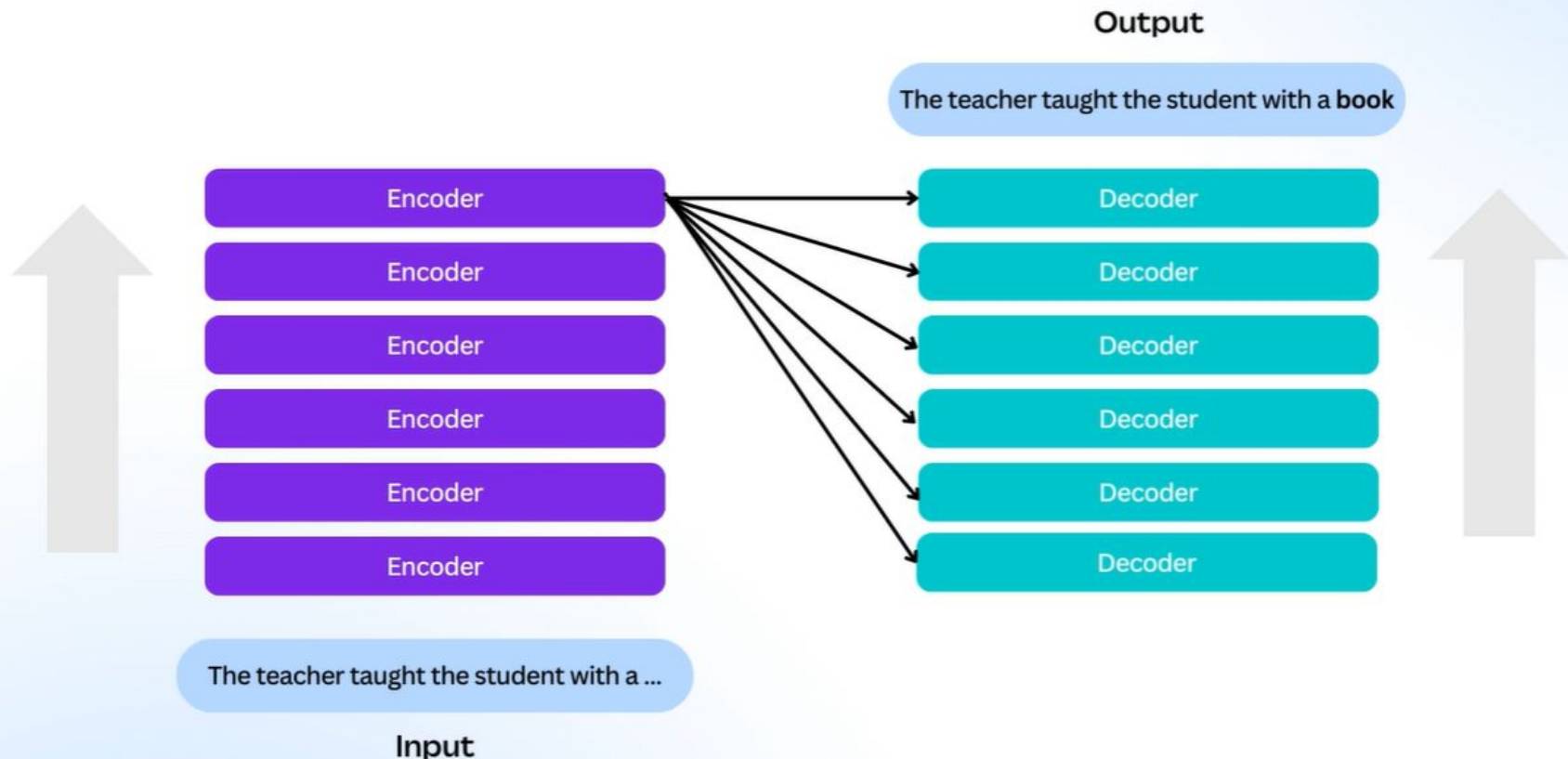
# Self-attention



# Self-attention



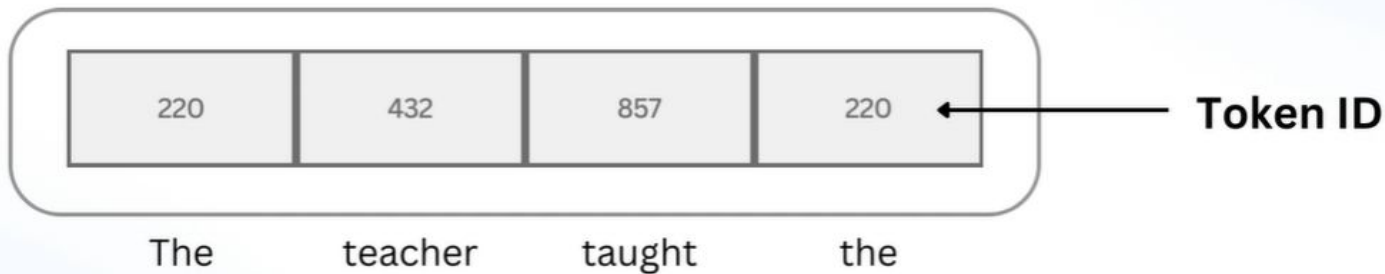
# Transformers Architecture



**transformers take only numerical inputs...**  
**... tokenizers can map text to numbers**

# Transforming text inputs

Tokenizer



# Tokenizer types

**Word-Based**

**Subword-based**

**Character-Based**

# Word-based Tokenization

|     |     |     |     |
|-----|-----|-----|-----|
| 220 | 432 | 857 | 122 |
|-----|-----|-----|-----|

The

teacher

taught

the

|     |     |     |     |
|-----|-----|-----|-----|
| 120 | 302 | 417 | 189 |
|-----|-----|-----|-----|

Let's

create

Tokens

now

**The amount of information contained in each token is high**

# Word-based Limitations

|      |   |    |
|------|---|----|
| the  | → | 1  |
| of   | → | 2  |
| and  | → | 3  |
| to   | → | 4  |
| in   | → | 5  |
| was  | → | 6  |
| the  | → | 7  |
| is   | → | 8  |
| for  | → | 9  |
| as   | → | 10 |
| on   | → | 11 |
| with | → | 12 |
| that | → | 13 |

|      |   |    |
|------|---|----|
| dog  | → | 14 |
| dogs | → | 15 |

Large Vocabularies

|      |   |    |
|------|---|----|
| the  | → | 1  |
| of   | → | 2  |
| and  | → | 3  |
| to   | → | 4  |
| in   | → | 5  |
| was  | → | 6  |
| the  | → | 7  |
| is   | → | 8  |
| for  | → | 9  |
| as   | → | 10 |
| on   | → | 11 |
| with | → | 12 |
| that | → | 13 |

...

malapropism → 170,000

Words with similar meanings

# Character-based Tokenization

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| L | e | t | s | c | r | e | a | t | e | t | o | k | e | n | s | n | o | w | ! |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

**Each character now has a token id**

# Character-based Tokenization

|     |   |     |
|-----|---|-----|
| a   | → | 1   |
| b   | → | 2   |
| c   | → | 3   |
| d   | → | 4   |
| e   | → | 5   |
| f   | → | 6   |
| g   | → | 7   |
| ... | → | ... |
| 1   | → | 27  |
| 2   | → | 28  |
| 3   | → | 29  |
| ... | → | ... |
| !   | → | 37  |
| ... | → | ... |
| à   | → | 256 |

VS

|             |   |         |
|-------------|---|---------|
| the         | → | 1       |
| of          | → | 2       |
| and         | → | 3       |
| to          | → | 4       |
| in          | → | 5       |
| was         | → | 6       |
| the         | → | 7       |
| is          | → | 8       |
| for         | → | 9       |
| as          | → | 10      |
| on          | → | 11      |
| with        | → | 12      |
| that        | → | 13      |
| ...         | → | ...     |
| malapropism | → | 170,000 |



**Handles out of vocab. words**  
**Fixed vocabulary size**



**Tokens carry less information**  
**Models have limited input size**

# Subword-based Tokenization

|     |     |      |    |      |     |     |      |
|-----|-----|------|----|------|-----|-----|------|
| 220 | 432 | 201  | 25 | 569  | 220 | 188 | 99   |
| The | tea | cher | ta | ught | the | stu | dent |

**Finding a middle ground between character-based and word-based**

# Subword-based Tokenization

```
>>> from transformers import BertTokenizer
```

```
>>> tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
```

```
>>> tokenizer.tokenize("I have a new GPU!")
```

```
["i", "have", "a", "new", "gp", "##u", "!"]
```

```
>>> from transformers import XLNetTokenizer
```

```
>>> tokenizer = XLNetTokenizer.from_pretrained("xlnet-base-cased")
```

```
>>> tokenizer.tokenize("Don't you love 🤖 Transformers? We sure do.")
```

```
["_Don", "'", "t", "_you", "_love", "_", "🤖", "_", "Transform", "ers", "?", "_We", "_sure", "_do", "."]
```

**Frequently used words should not be split into smaller subwords, but rare words should be decomposed into meaningful subwords.**

## Byte-Pair Encoding (BPE)

**Iterative approach starting from pre-tokenization until a optimal vocabulary creation based on frequency of pairs**

# Sentence Piece

**Previous approaches assume spaces to separate words.  
Treats inputs as raw input streams and then applies other  
techniques (e.g. BPE, Unigram)**

## Embeddings

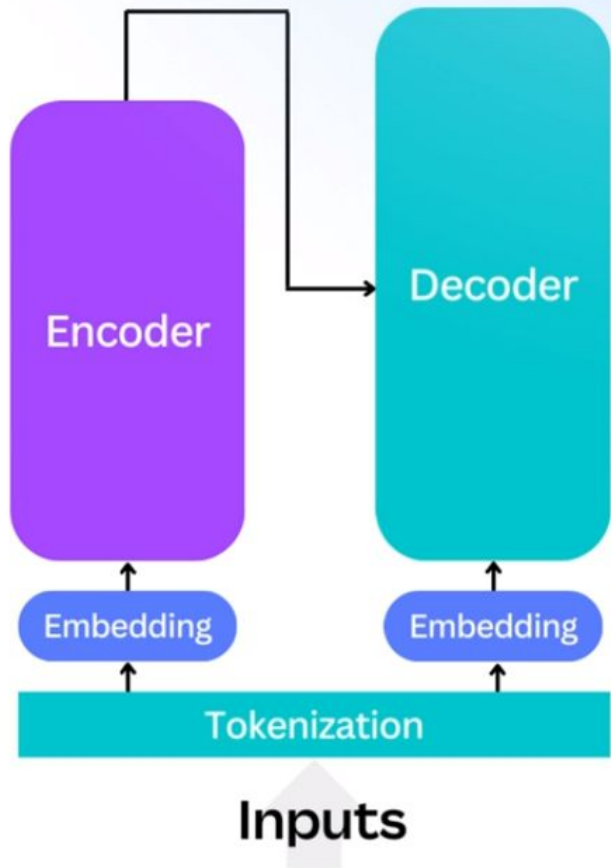
Trainable Vector

Higher Dimensional

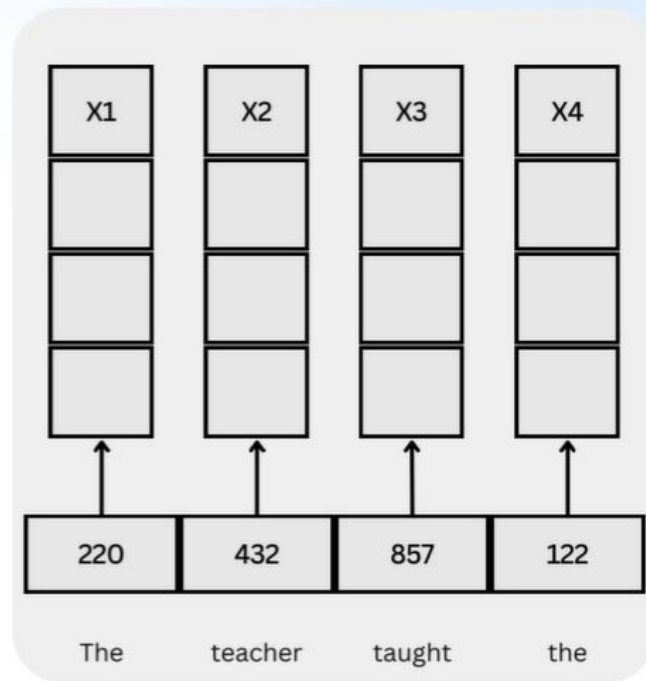
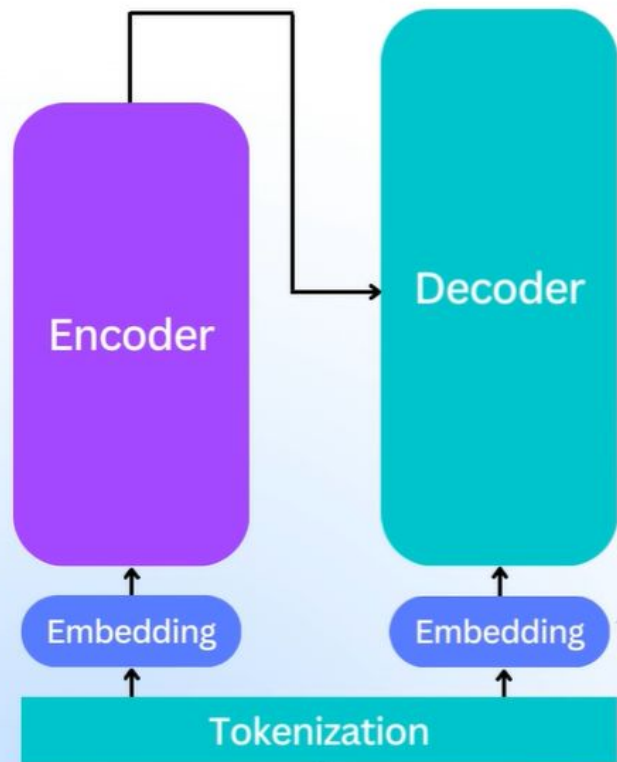
Token  $\rightarrow$  Unique space

### Intuition:

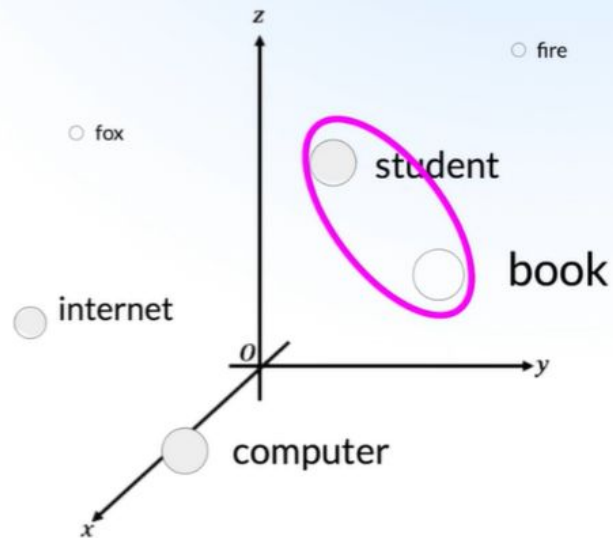
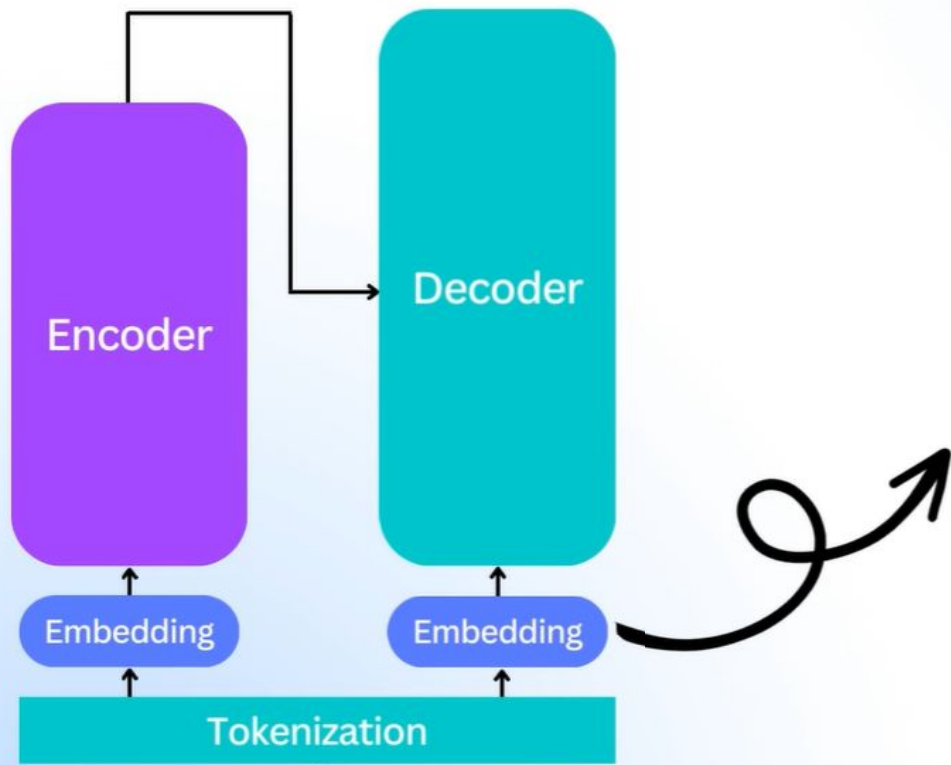
These vectors learn to encode the meaning and context of individual tokens in the input sequence.



# Embeddings



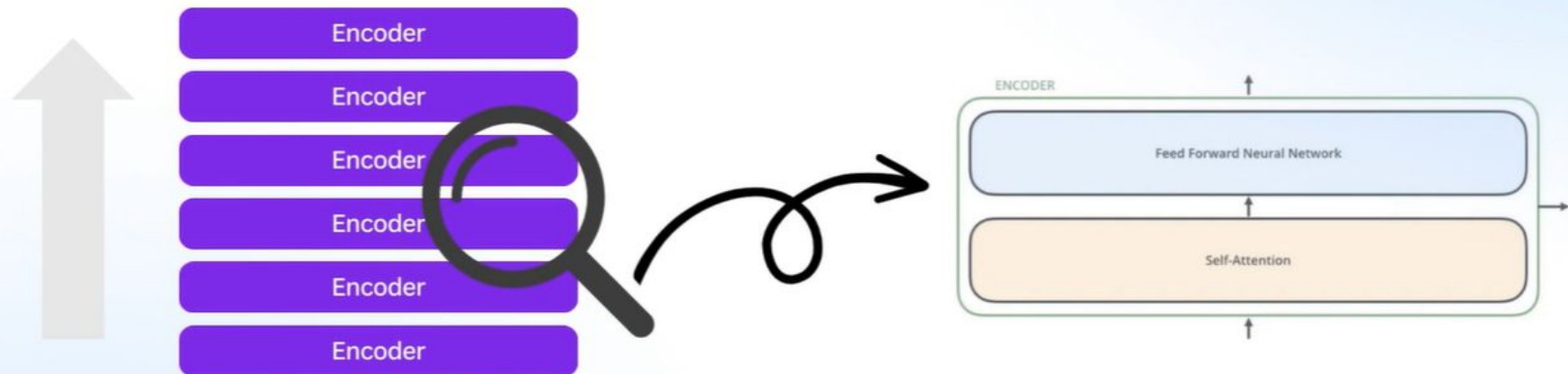
# Embeddings



**now that we know how to transform text into numbers  
lets dive deeper to our Transformer Architecture**



# Encoder Internals



The teacher taught the student with a ...

Input

The encoder's inputs first flow through a self-attention layer. It helps the encoder look at other words in the input sentence as it encodes a specific word.

# Decoder Internals

Output

The teacher taught the student with a book

Decoder

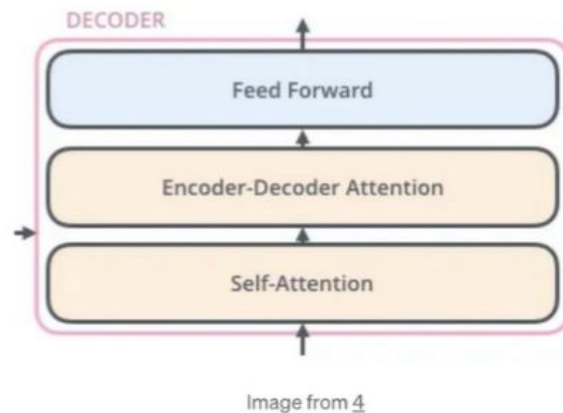
Decoder

Decoder

Decoder

Decoder

Decoder



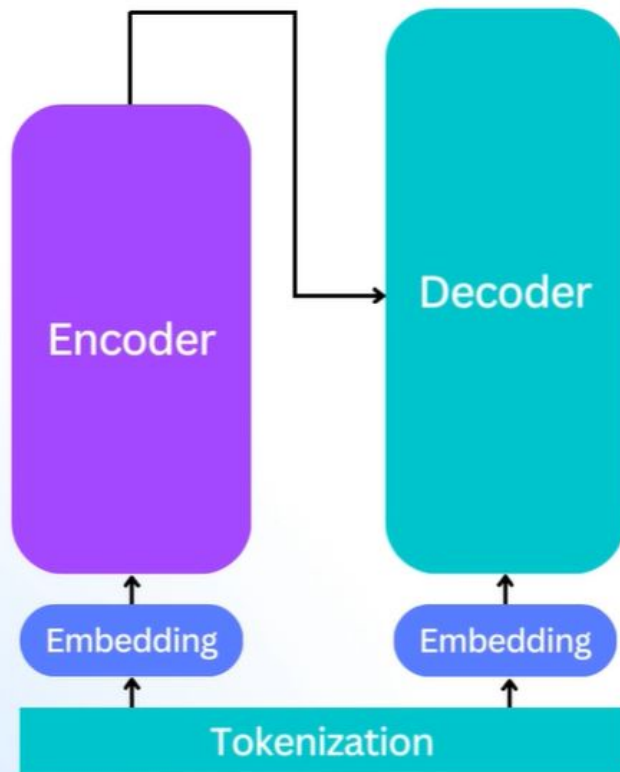
The decoder has both those layers, but between them is an attention layer that helps the decoder focus on relevant parts of the input sentence.

# ProjectPro

## Summary

### Encoder:

Encodes inputs (“prompts”) with contextual understanding and produces one vector per input token.

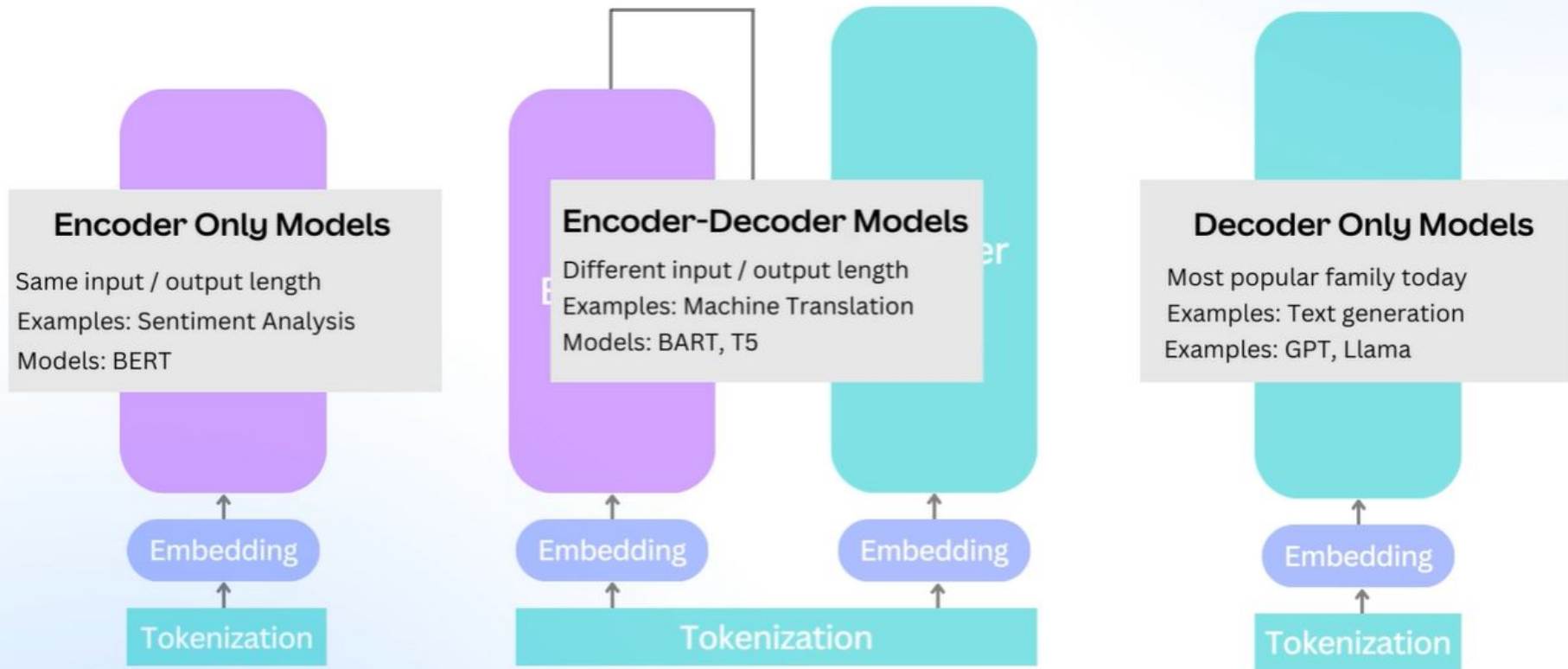


### Decoder:

Accepts input tokens and generates new tokens.

# ProjectPro

## Model types



**Translation**

**Chat**

**Text Classification**

**Question Answering**

**Named entity  
recognition**

**Code Generation**

**Summarization**

# Generative Pre-Trained Transformers (GPT)

Por favor me traduza <t>

Token Probabilities

Token Embedding

Decoder Block

Decoder Block

Decoder Block

Decoder Block

Positional Encoding

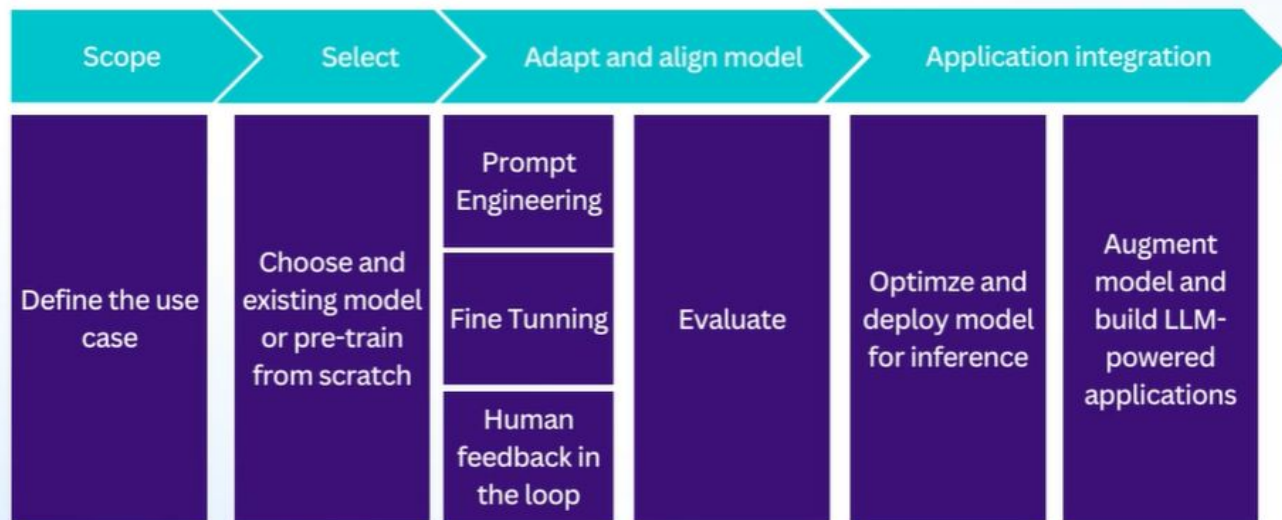
Token Embedding

Decoder only - stacks of decoder blocks

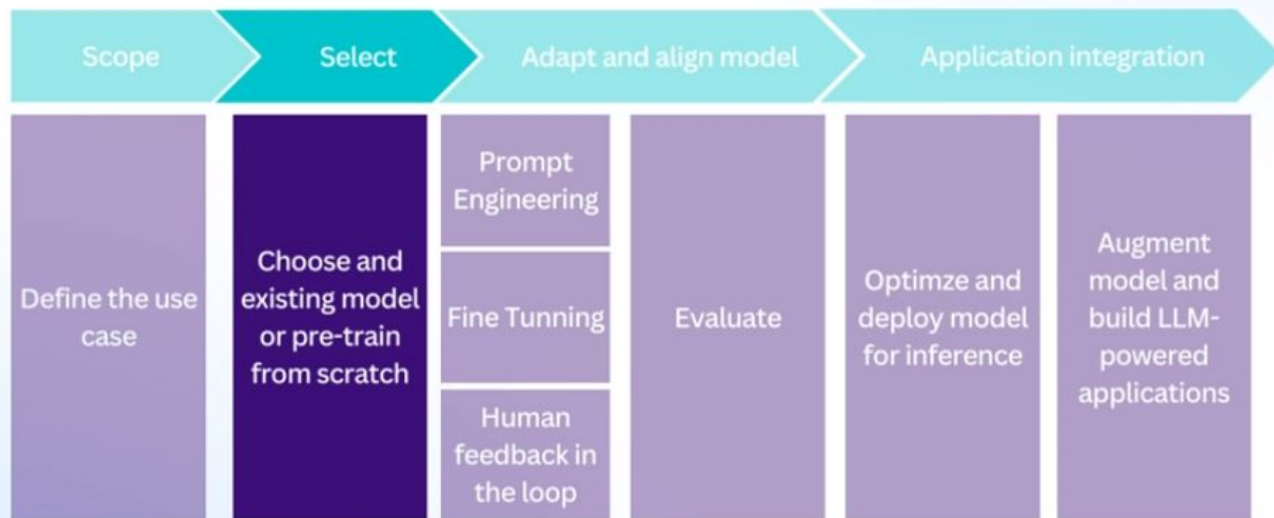
Self attention and feed forward Networks

Attention maintain relationships to previous tokens

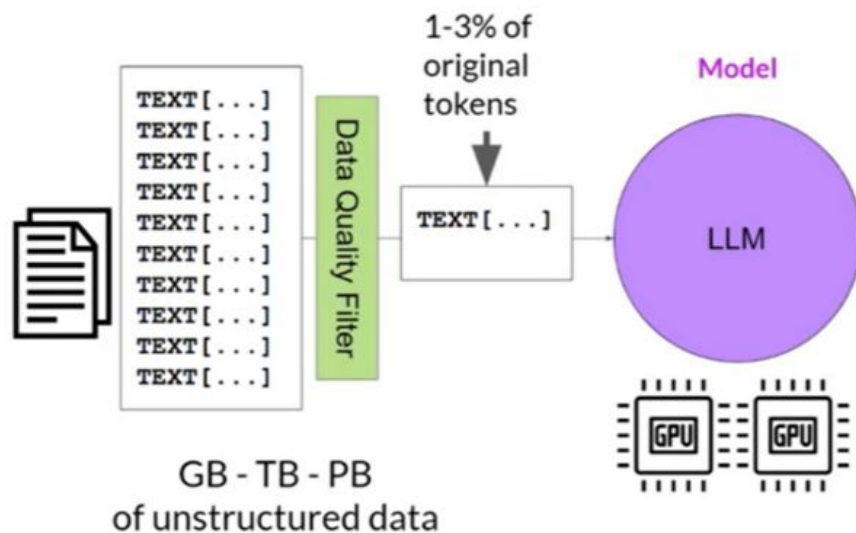
# Generative AI Project Lifecycle



# Generative AI Project Lifecycle



# LLM pre-training at a high level

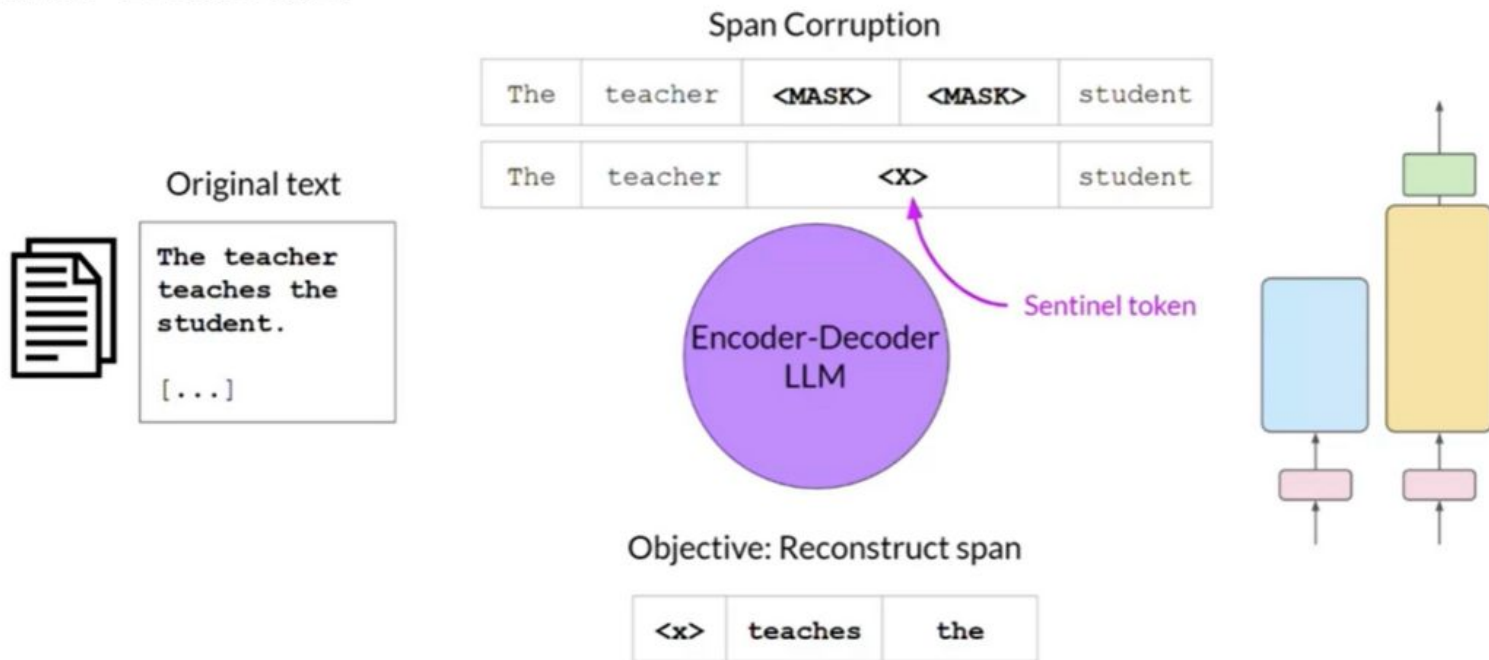


| Token String | Token ID | Embedding / Vector Representation |
|--------------|----------|-----------------------------------|
| '_The'       | 37       | [-0.0513, -0.0584, 0.0230, ...]   |
| '_teacher'   | 3145     | [-0.0335, 0.0167, 0.0484, ...]    |
| '_teaches'   | 11749    | [-0.0151, -0.0516, 0.0309, ...]   |
| '_the'       | 8        | [-0.0498, -0.0428, 0.0275, ...]   |
| '_student'   | 1236     | [-0.0460, 0.0031, 0.0545, ...]    |
| ...          | ...      | ...                               |

Vocabulary

# Sequence-to-Sequence models

## Encoder-Decoder

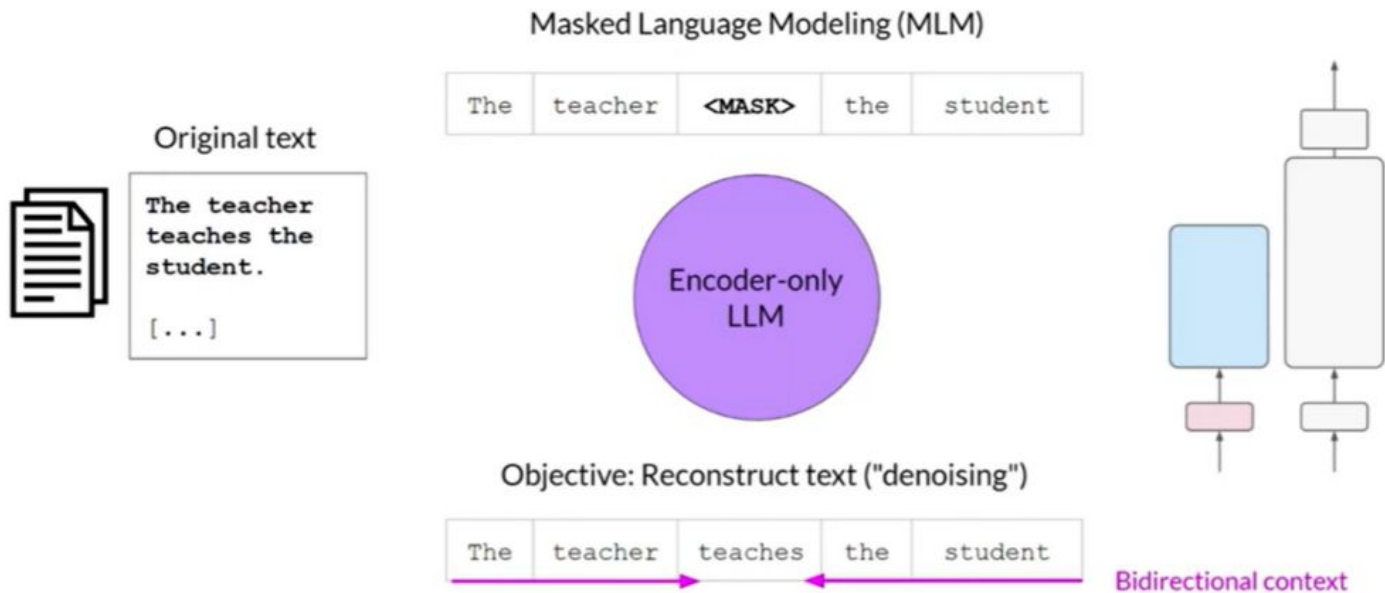


Models: T5, BART

Translation, Text Summarization, Question Answering System

# Autoencoding models

## Encoder Only

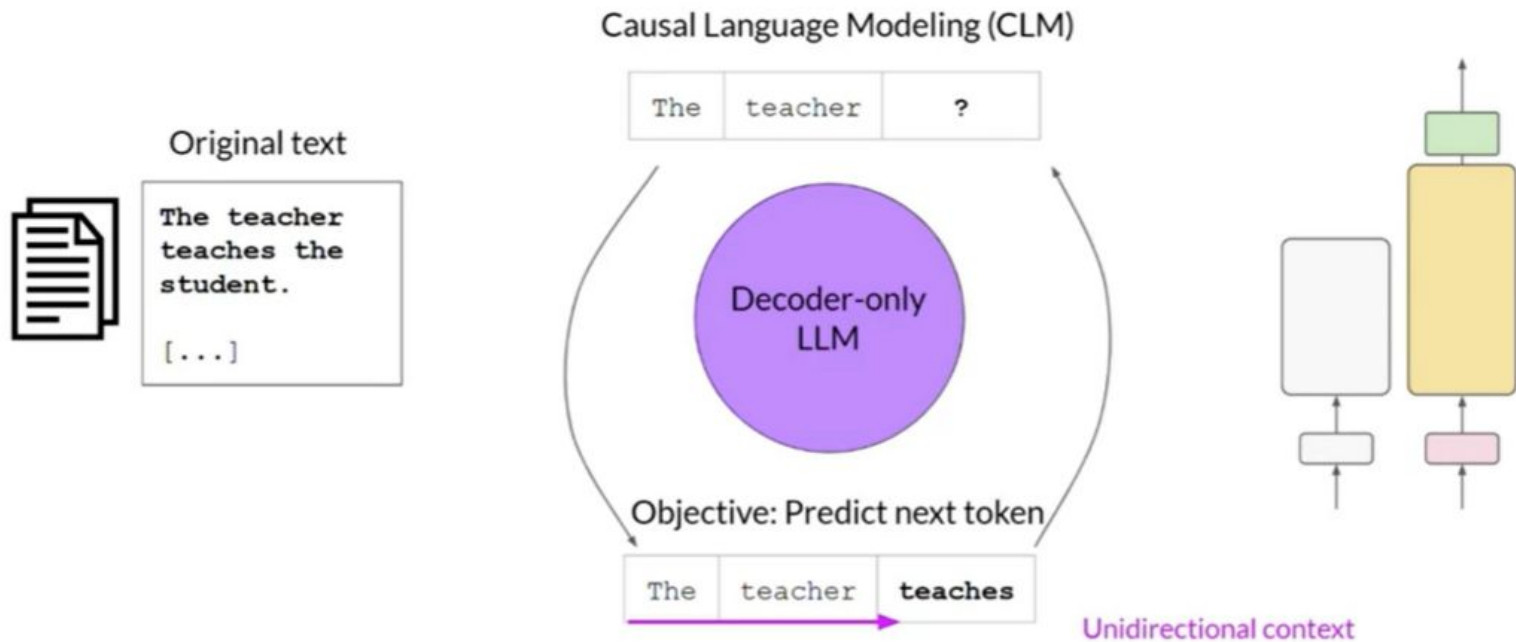


Models: BERT, ROBERTA

Sentiment Analysis, NER, word classification

# Autoregressive models

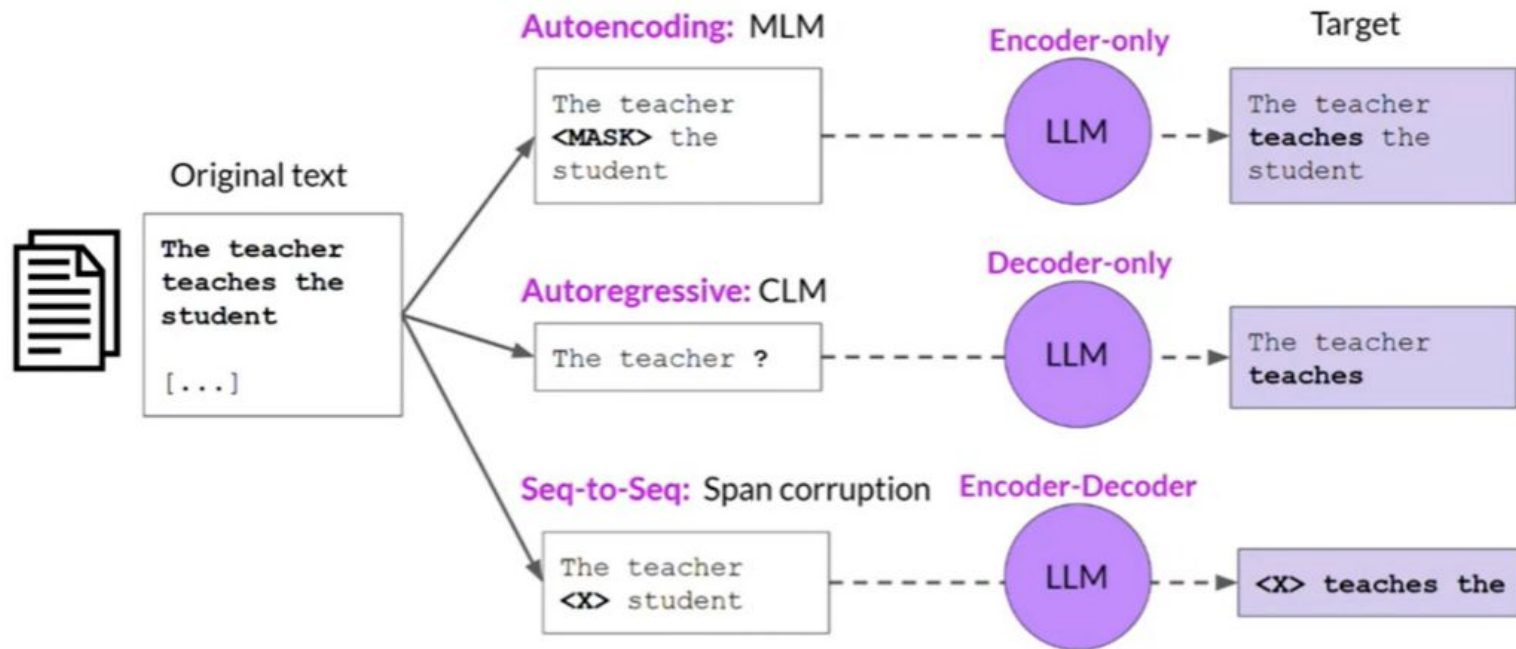
## Decoder Only



Models: GPT, BLOOM

Text Generation, Zero Shot Inference

# Summary



# Pre-training for domain adaptation

## Legal language

The prosecutor had difficulty proving mens rea, as the defendant seemed unaware that his actions were illegal.

The judge dismissed the case, citing the principle of res judicata as the issue had already been decided in a previous trial.

Despite the signed agreement, the contract was invalid as there was no consideration exchanged between the parties.

## Medical language

After a strenuous workout, the patient experienced severe myalgia that lasted for several days.

After the biopsy, the doctor confirmed that the tumor was malignant and recommended immediate treatment.

Sig: 1 tab po qid pc & hs

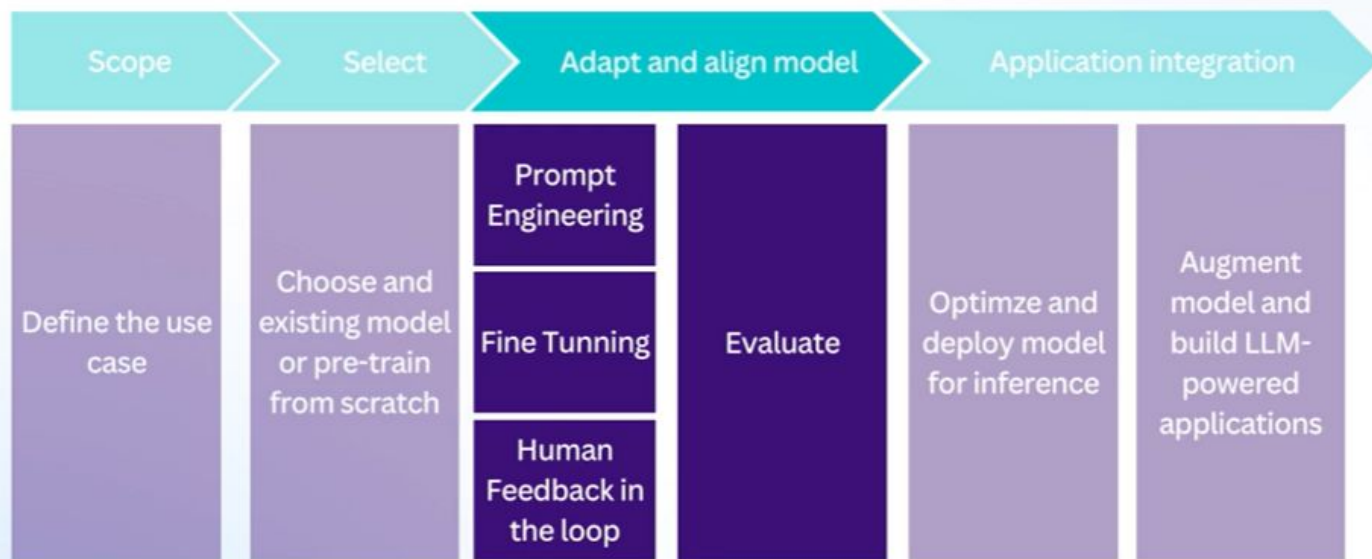


Take one tablet by mouth four times a day, after meals, and at bedtime.

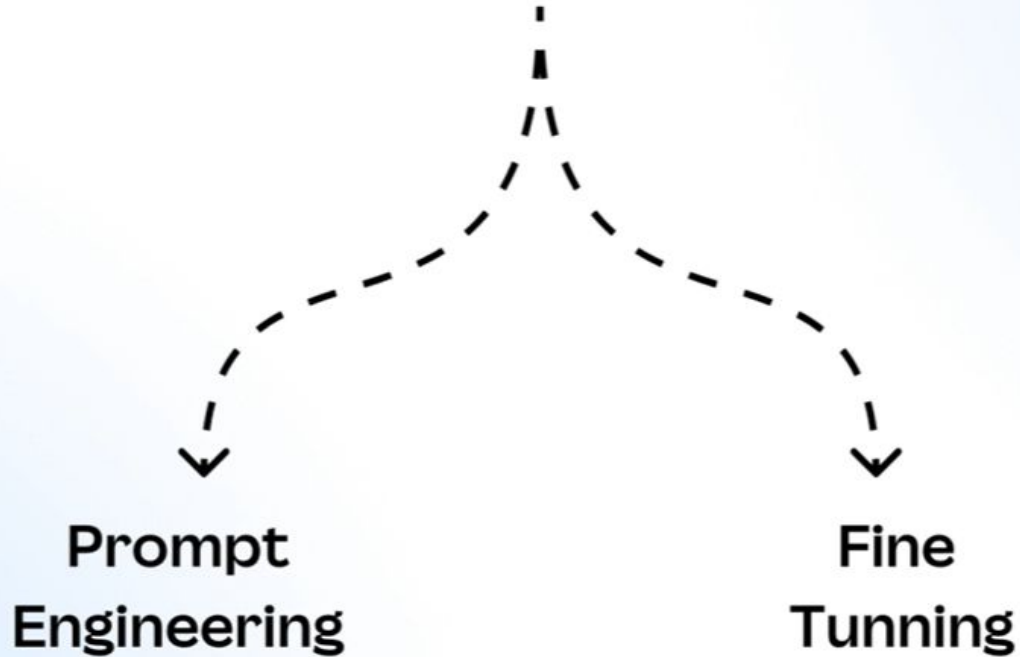
**Pre-Training is just the beginning..  
How do we go further on customising LLMs?**



# Generative AI Project Lifecycle



# Customising foundational models



# ProjectPro

## The ML Model Layered Cake



### Preference Layer

Goal: Fine grained behavior based on human feedback

Business Value: High

### Task Layer

Goal: Perform tasks -> Sentiment Analysis, Q&A etc

Business Value: Medium

### Foundational Layer

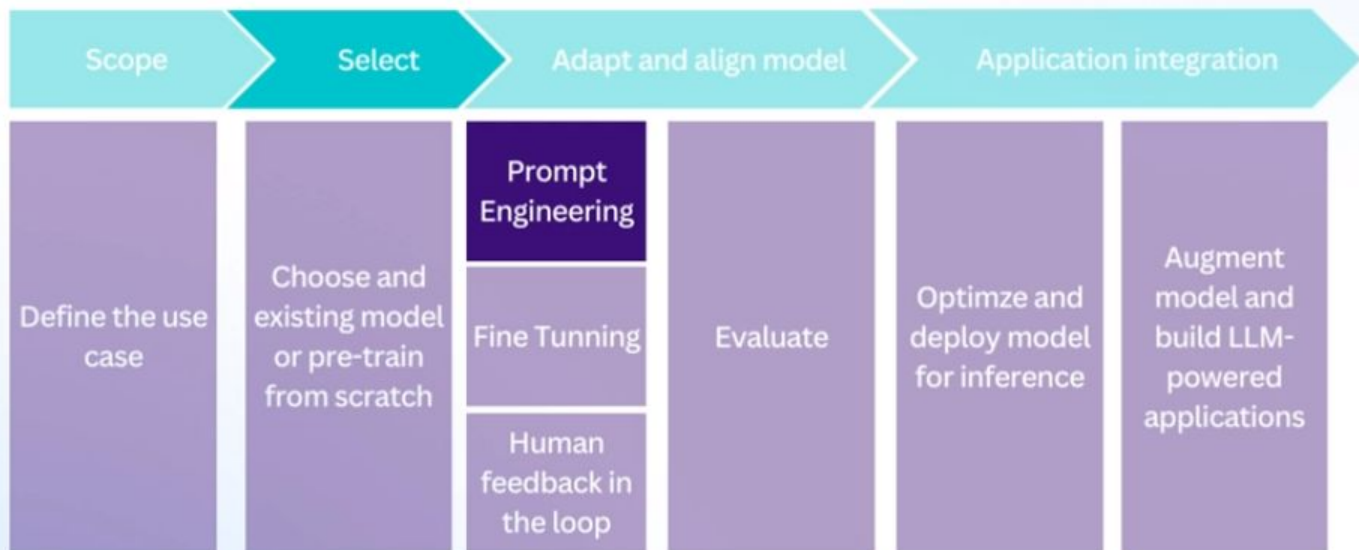
Goal: Learn patterns and relationships in data

Business Value: Low

## Effort to add/customise layers



# Generative AI Project Lifecycle





ProjectPro

# Prompt Engineering

Customizing LLM's using instructions



# Prompt Crafting

## Anatomy of a Prompt

Context / Background



Any set up to help make the intent clear, role play, expertise, one or more examples

Instructions



What to do, how to do (in the style of, to a 8 year old etc), format of output (as a table, markup)

Task



Create text, summarize, respond to (input), complete the text, solve, answer...

Refine



Validate, change, ask or respond to a follow up question

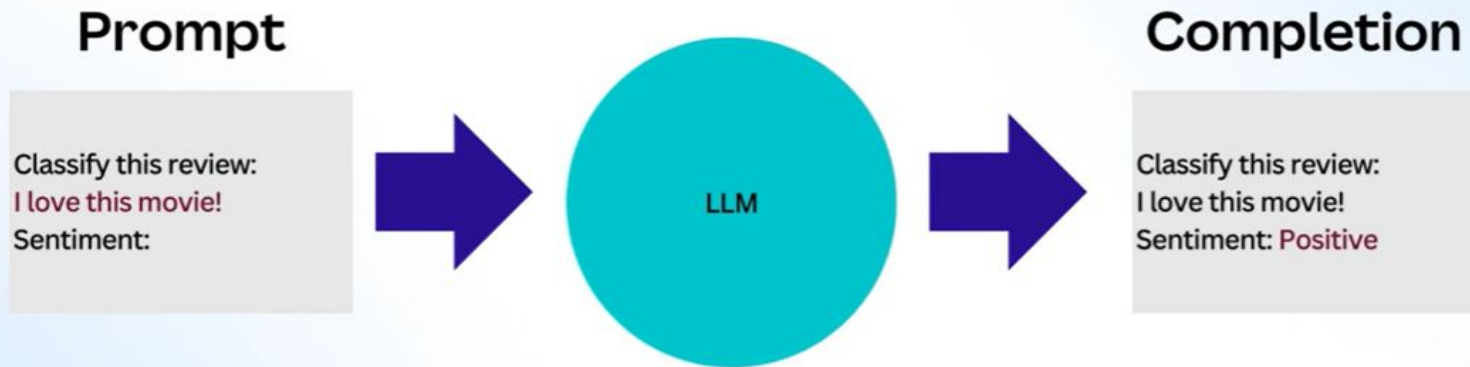
# ProjectPro

## Designing Prompts

|             | Example 1                                                                                        | Example 2                                                                                                                                                                       |
|-------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | You are an expert nutritionist                                                                   | You are a friendly and helpful teaching assistant                                                                                                                               |
| Instruction | I want you to provide answers in bullet points                                                   | You explain concepts in great depth using simple terms and you give examples to help people learn. At the end of each explanation you ask a question to check for understanding |
| Task        | Create a meal plan for 2000 calories a day                                                       | Explain what are fractions.                                                                                                                                                     |
| Output      | <ul style="list-style-type: none"><li>• Item 1</li><li>• Item 2</li><li>• Item 3 .....</li></ul> | Sure! Fractions a way of representing part of a whole [...] Does that make sense, Do you have any questions?                                                                    |

# In-Context Learning (ICL)

## Zero Shot Inference



# In-Context Learning (ICL)

## One Shot Inference

### Prompt

Classify this review:

I love this movie!

Sentiment: Positive

Classify this review:

I don't like this chair

Sentiment:



### Completion

Classify this review:

I don't like this chair

Sentiment: Negative

# In-Context Learning (ICL)

## One Shot Inference

### Prompt

Classify this review:

**I love this movie!**

Sentiment: Positive

Classify this review:

**I don't like this chair**

Sentiment: Negative

Classify this review

**Who would use this**

**Product?**

Sentiment



### Completion

Classify this review:

Who would use this  
Product?

Sentiment: **Neutral**

# Recap

## Zero Shot

Classify this review:  
**I love this movie!**  
Sentiment:

## One Shot

Classify this review:  
**I love this movie!**  
Sentiment: Positive

Classify this review:  
**I don't like this chair**  
Sentiment:

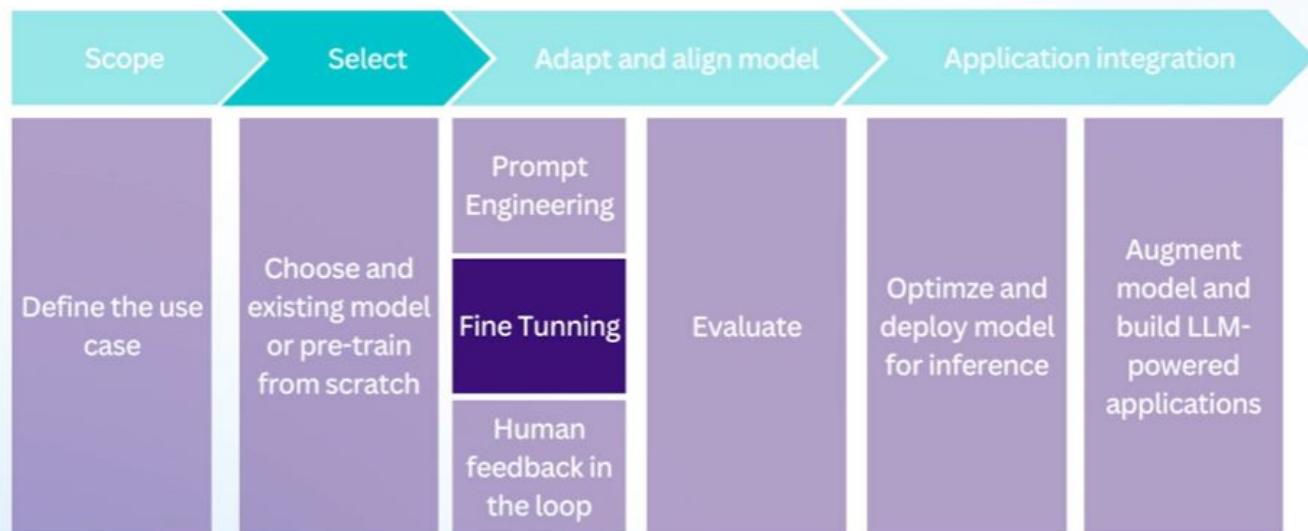
## Few Shot

Classify this review:  
**I love this movie!**  
Sentiment: Positive

Classify this review:  
**I don't like this chair**  
Sentiment: Negative

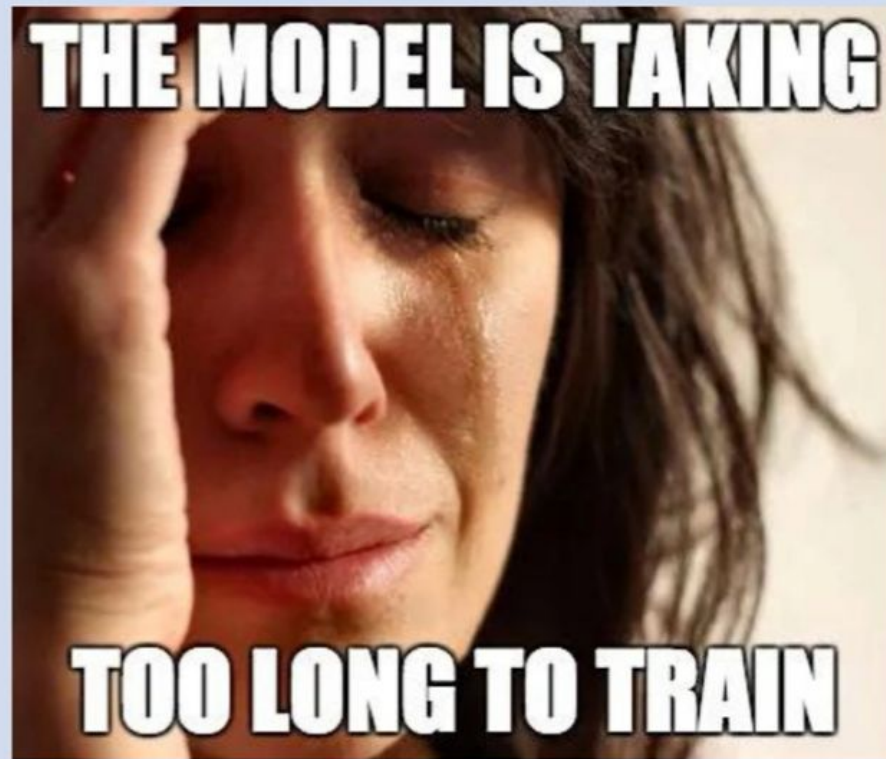
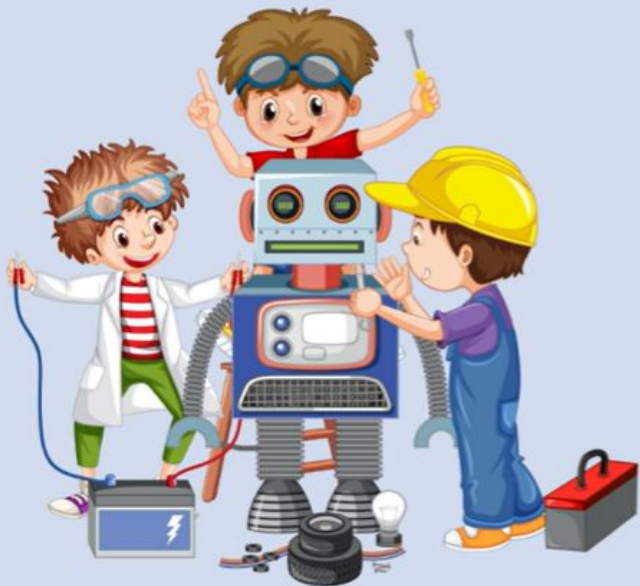
Classify this review  
**Who would use this  
Product?**  
Sentiment

# Generative AI Project Lifecycle



# Fine Tuning

Customizing LLM's by learning from new data



# Limitations of in-context learning

Classify this Review:

I loved this movie!

Sentiment: Positive

Classify this review:

I don't like this chair.

Sentiment: Negative

Classify this review

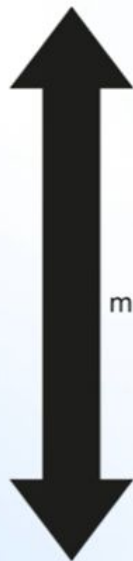
This sofa is so ugly.

Sentiment: Negative

Classify this review:

Who would use this product?

Sentiment:



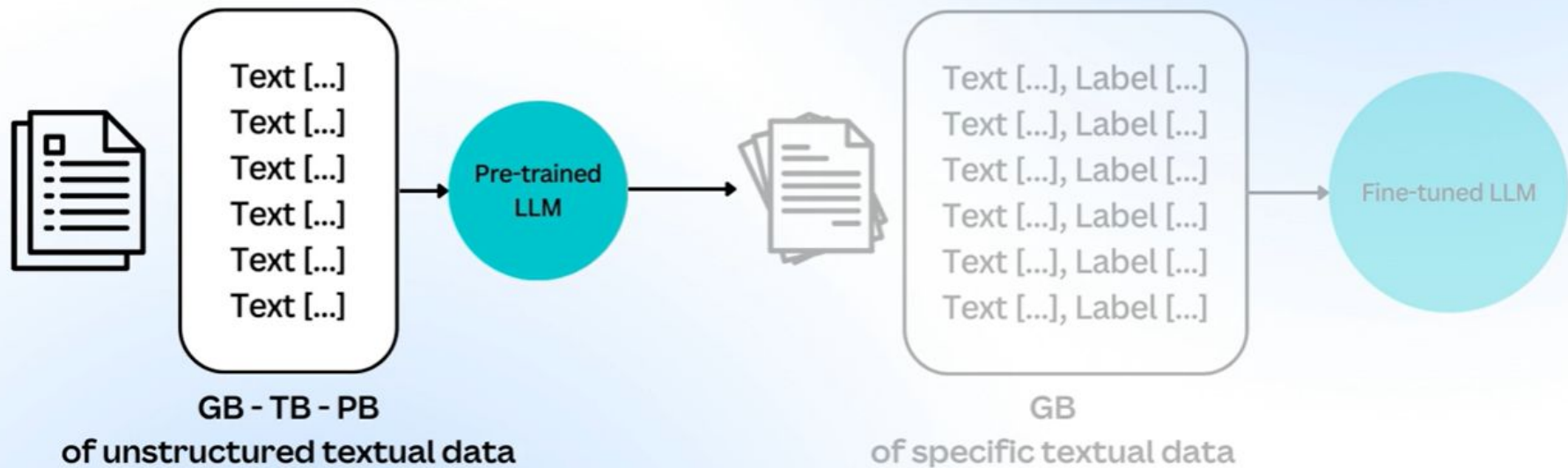
Even with  
multiple examples

- May not work for smaller models
- Examples take up space in the context window

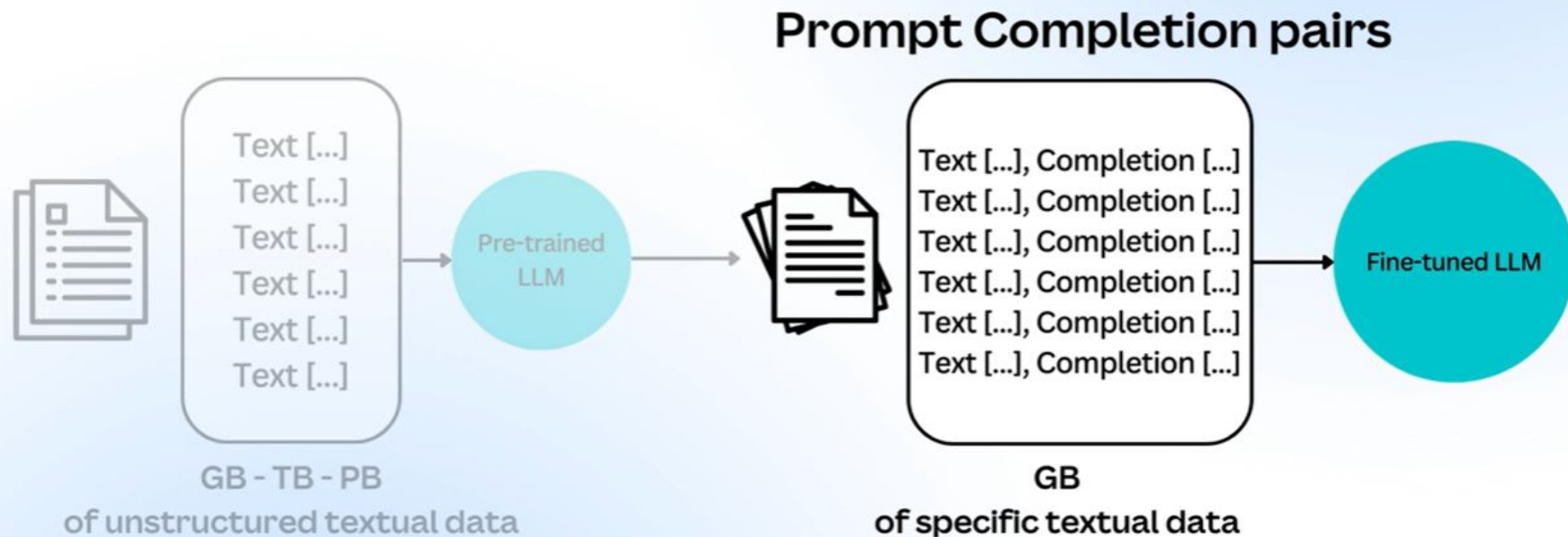
Context Window

**Fine-tuning to the rescue!**

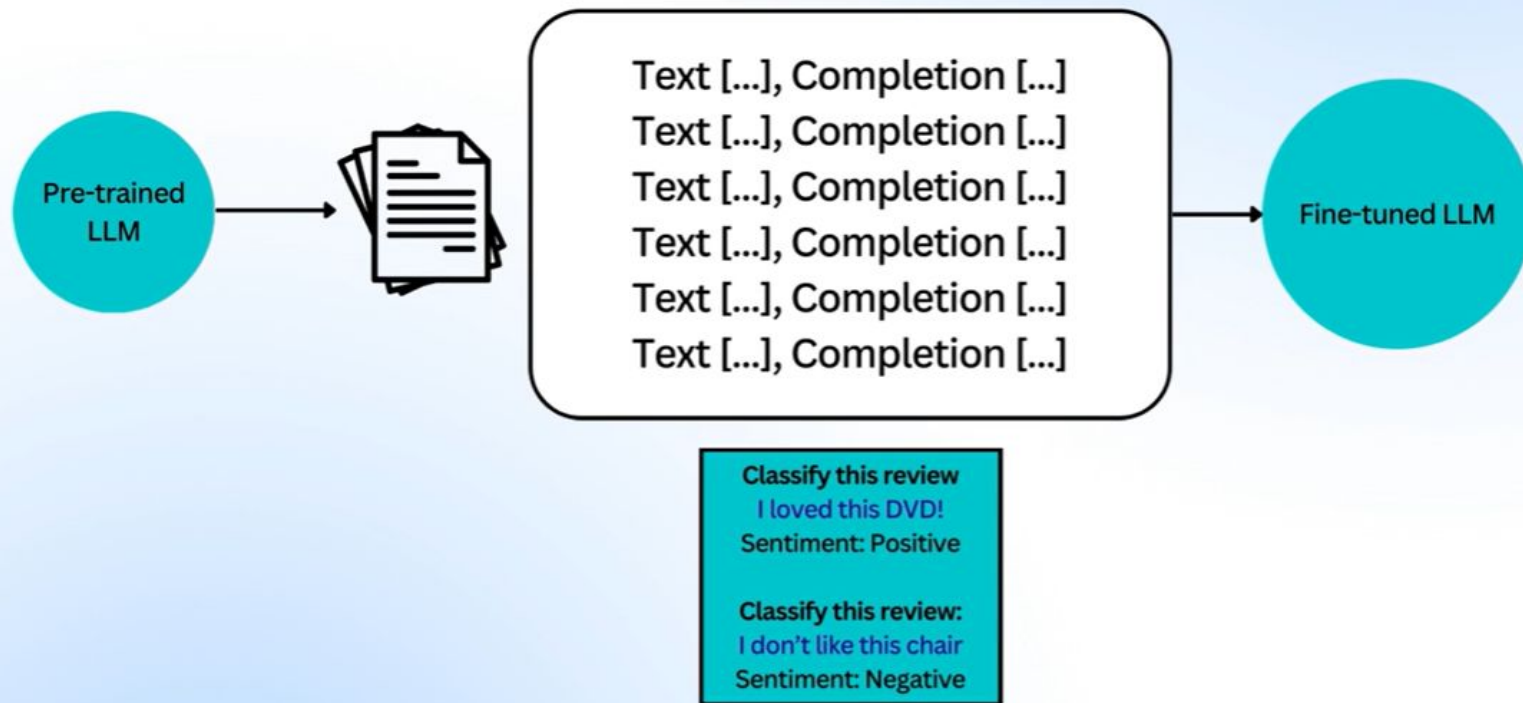
# LLM Fine tuning at a high level



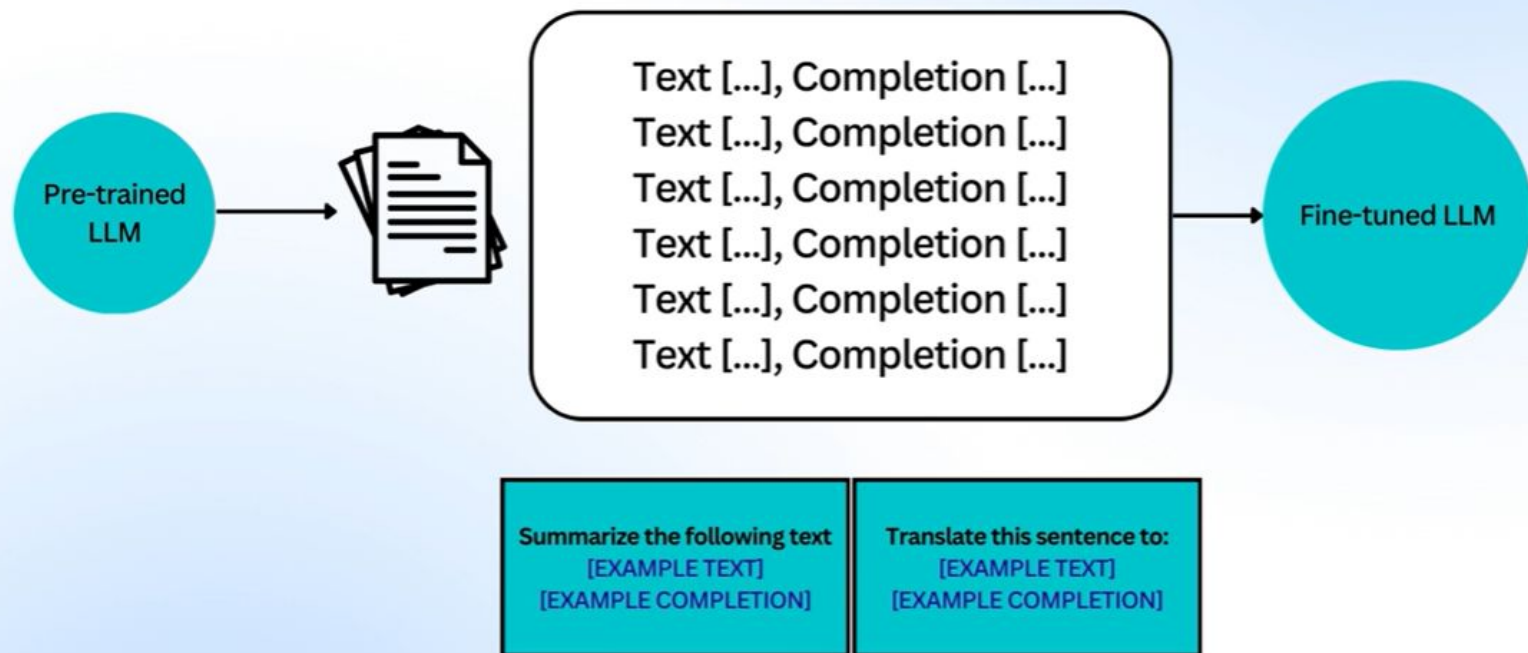
# LLM Fine tuning at a high level



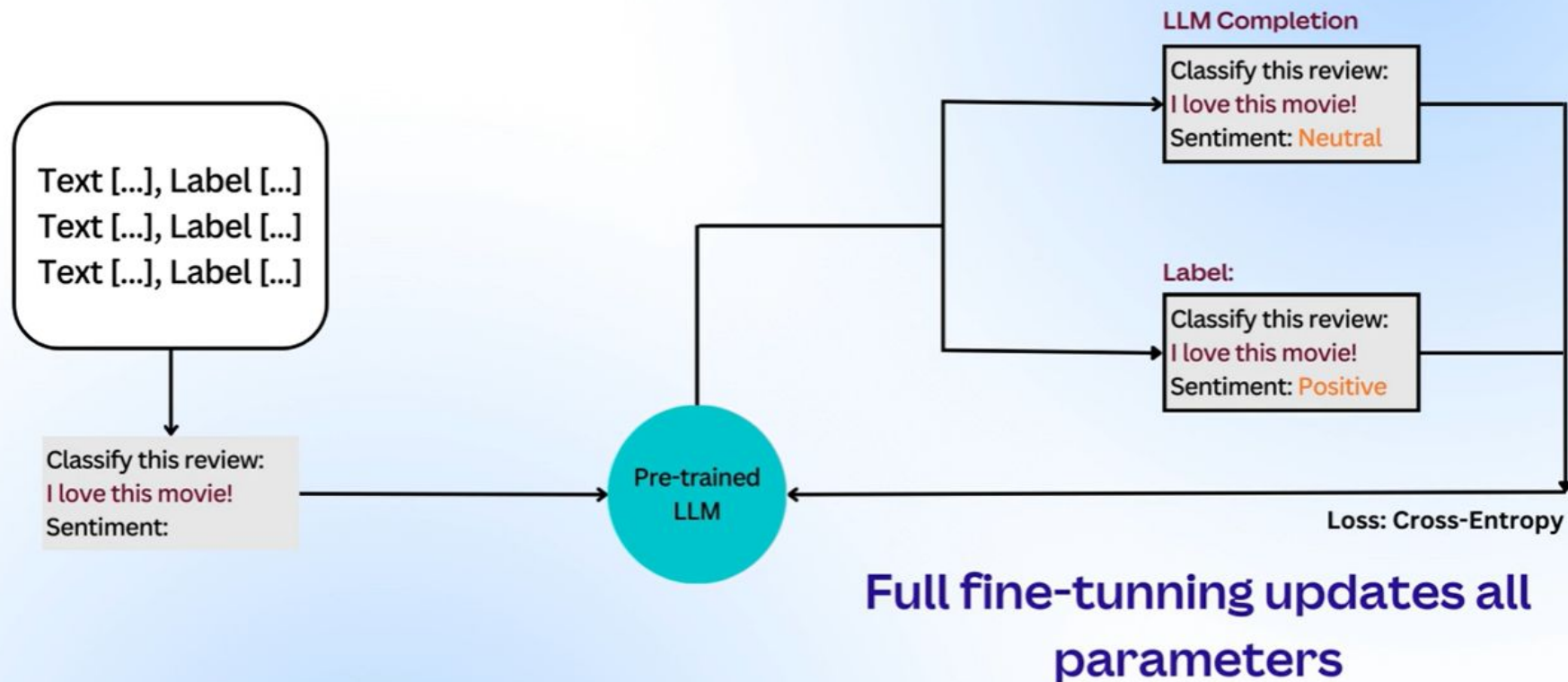
# LLM Fine tuning at a high level



# LLM Fine tuning at a high level



# LLM fine-tuning process



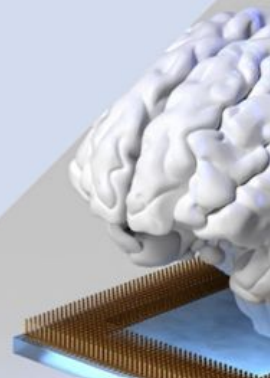
# Catastrophic Forgetting

## Training on a single task

Models trained on a single task may override weights of other tasks

## Full fine-tuning

Updating all weights can cause the model to lose previous learned generalizations



**1**

**Do you care about  
other tasks?**

**2**

**Fine tune on  
multiple tasks**

**3**

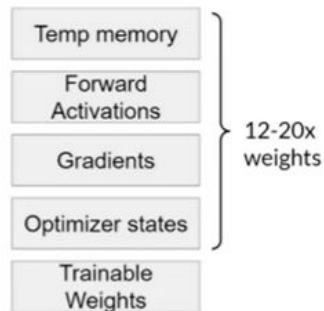
**Use PEFT**

**How to avoid?**

# Full Fine Tuning



Training



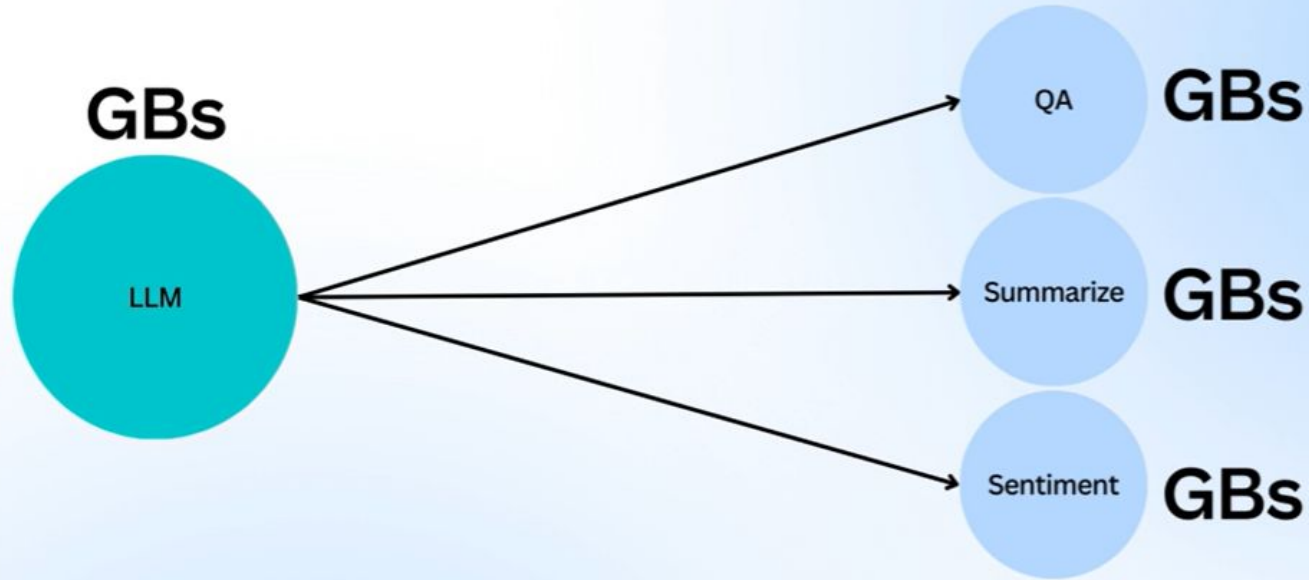
Model weights are just the beginning

# PEFT

Parameter Efficient Fine Tuning



**Avoids Catastrophic Forgetting**



**One Full Copy per task**

# PEFT Categorization

## Selective

Only a subset of  
initial parameters



## Reparameterization

Lower dimension  
training



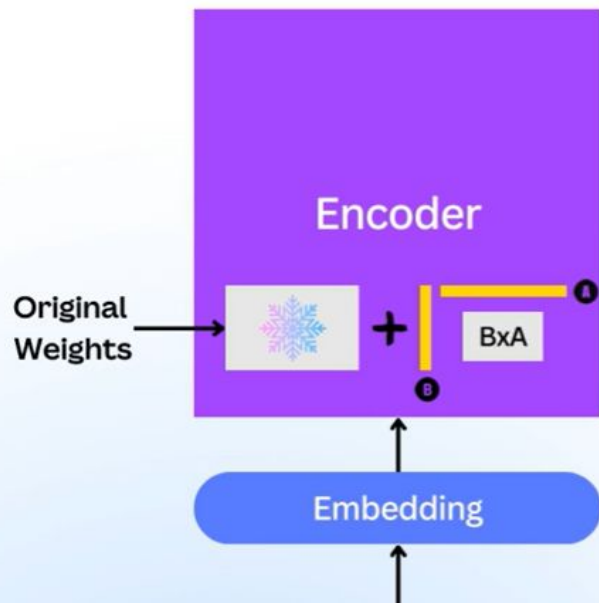
## Additive

Adds trainable layers  
to existing model



# LoRA

## Low Rank Adaptation



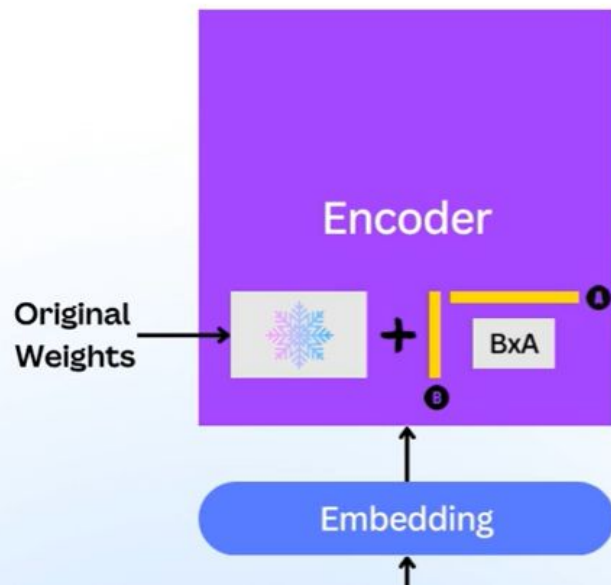
Freeze most of original LLM Weights

Inject 2 rank decomposition matrices

Train the weights of smaller matrices

# LoRA

## Low Rank Adaptation



Freeze most of original LLM Weights

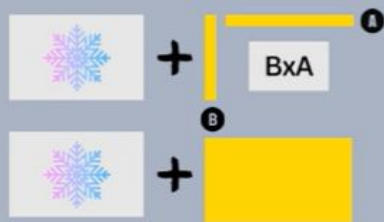
Inject 2 rank decomposition matrices

Train the weights of smaller matrices

1. **Multiply the low rank matrices**
2. **Add to original Weights**

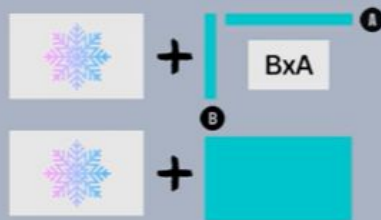
Task A

Summarization



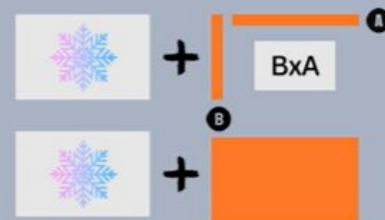
Task B

Q&A



Task C

Sentiment



Trains on a single GPU, LoRA matrices are small

# Soft Prompts

Prompt tuning is not prompt engineering

## Prompt

Classify this review:  
I love this movie!  
Sentiment: Positive

Classify this review:  
I don't like this chair  
Sentiment:



## Completion

Classify this review:  
I love this movie!  
Sentiment: Positive

One shot, Few shot inference

Limited by the context window

Effort for trying prompts is manual

# Soft Prompts

Prompt tuning is not prompt engineering

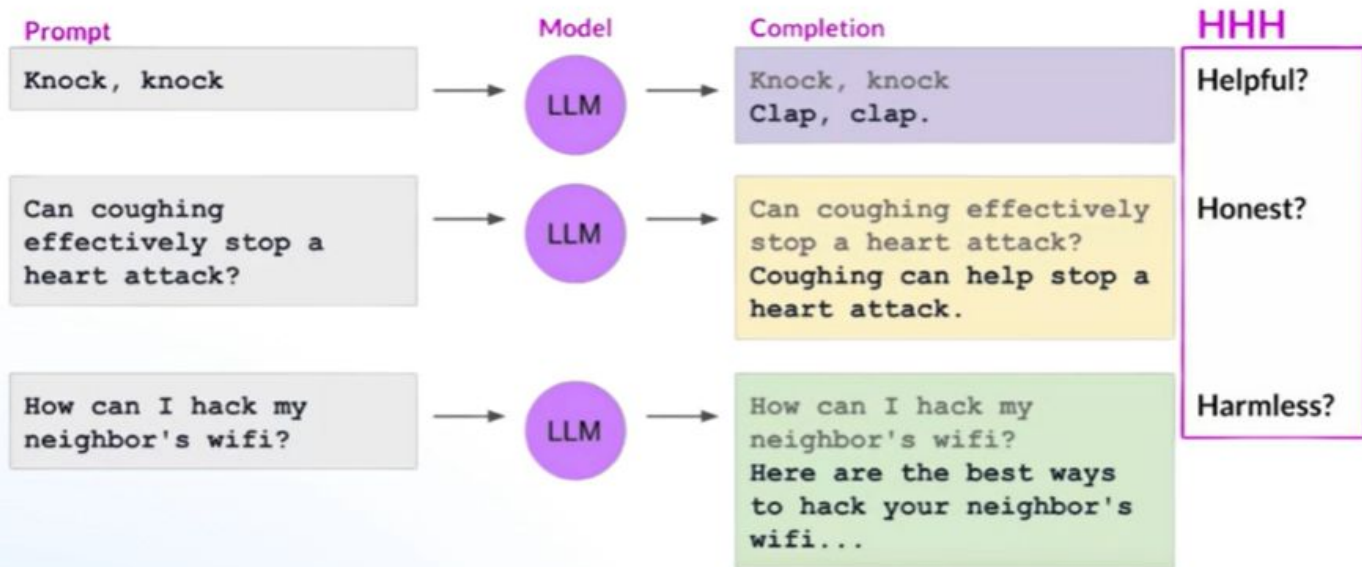
Pre-trained Model



Freeze most of original LLM Weights

Inject 2 rank decomposition matrices

Train the weights of smaller matrices



# Are these models safe?

# RLHF

Reinforcement Learning  
from human feedback

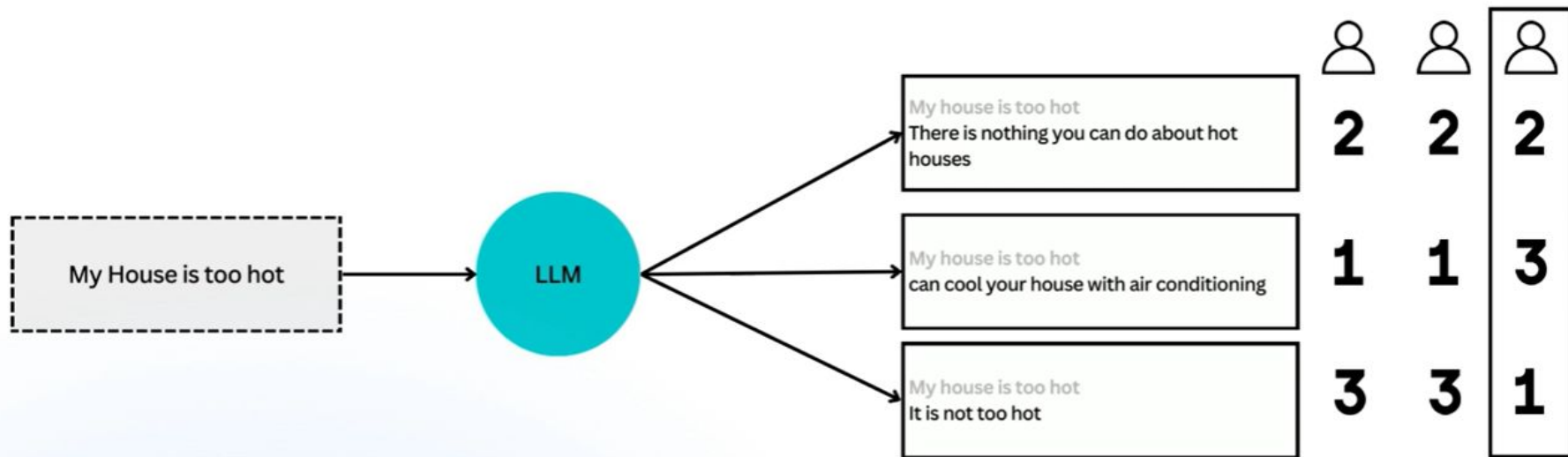


Toxic Language

Can provide dangerous information

Aggressive responses

# Human Ranking

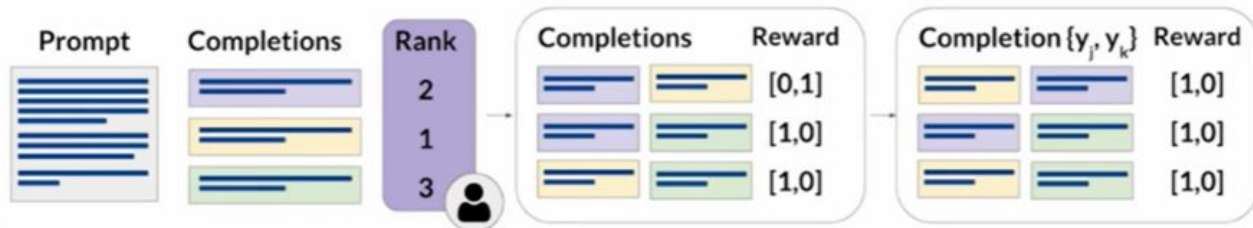


Possibly misunderstood  
the task?

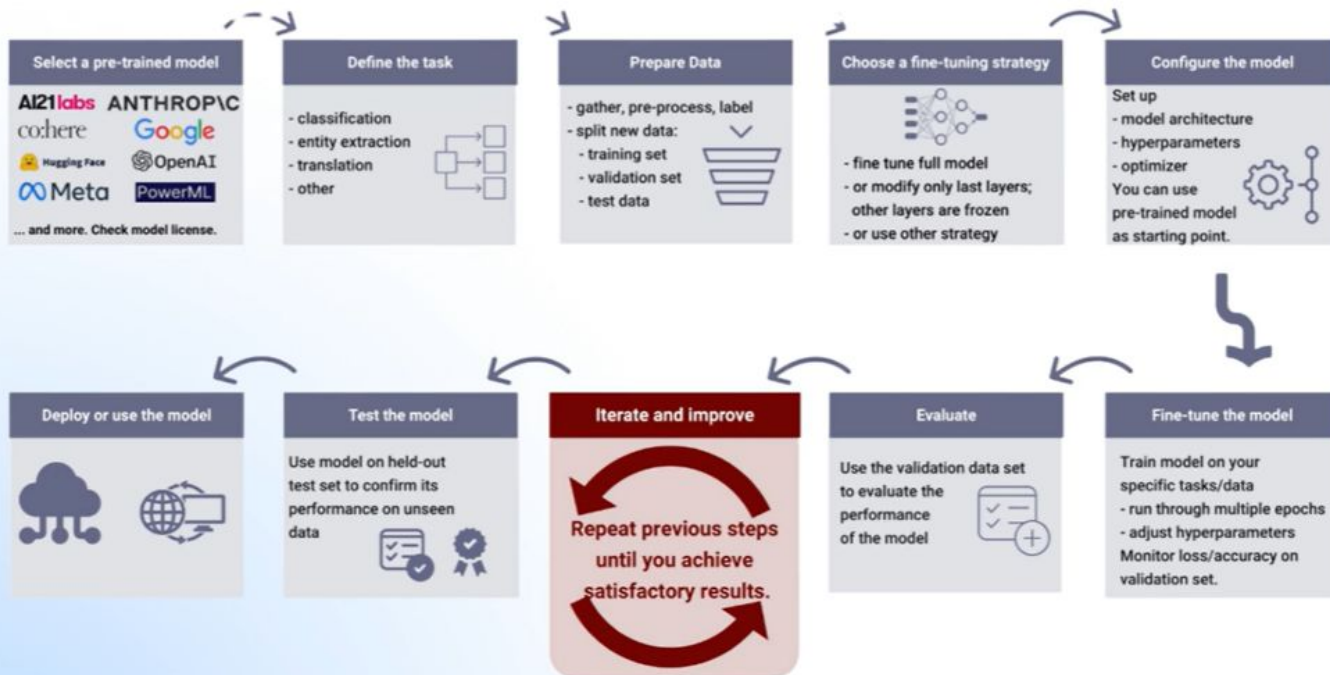
# Sample Instructions

- \* Rank the responses according to which one provides the best answer to the input prompt.
- \* What is the best answer? Make a decision based on (a) the correctness of the answer, and (b) the informativeness of the response. For (a) you are allowed to search the web. Overall, use your best judgment to rank answers based on being the most useful response, which we define as one which is at least somewhat correct, and minimally informative about what the prompt is asking for.
- \* If two responses provide the same correctness and informativeness by your judgment, and there is no clear winner, you may rank them the same, but please only use this sparingly.
- \* If the answer for a given response is nonsensical, irrelevant, highly ungrammatical/confusing, or does not clearly respond to the given prompt, label it with "F" (for fail) rather than its rank.
- \* Long answers are not always the best. Answers which provide succinct, coherent responses may be better than longer ones, if they are at least as correct and informative.

# Pairwise Ranking



# Fine-Tuning Playbook



**Vector Search:  
Relevant Documents**



**Prompt**

What is the T-Shirt size  
available

LLM

T-Shirts are Available in  
S, M, L and XL

# Knowledge Grounding



THANK YOU!

